

NOETHER: Constructive Metamorphic Pattern Identification from Operator Algebras and a Falsifiable Invariance-Blindness Theorem

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Metamorphic Testing relies on metamorphic relations (MRs), but MR identification remains experience- or search-driven. This paper studies MR identification for operator-algebraic program families rather than fault-detection superiority. We introduce **NOETHER**, a two-level framework for equation-governed or relational-semantics families whose behaviour admits an operator algebra. The upstream layer curates the family algebra and its structural components: symmetry, order, self-adjointness, time reversal, limits, qualitative dynamics, method comparison, and relational equivalence. The downstream CONSTRUCT-MP procedure derives five MetaPatterns and ten MR families from these components and checks whether a program interface can instantiate executable MRs. The construction is closed under its derivation operator, giving a well-formedness guarantee over the algebra-induced MR space. For the G and T^* MetaPatterns, an *Invariance-Blindness Theorem* characterises which implementation faults algebra-derived MRs can and cannot detect within the stated fault class. We evaluate NOETHER through four research questions covering canonical coverage, origin and boundary explanation, cross-domain and real-fault recurrence, and secondary executability and blind-spot checks. The evidence includes expert/search-derived MR corpora, three algebra instantiations, 21 in-scope real faults, and a 20-mutant equivariant-ML study. NOETHER identifies structural coverage not explicit in expert or search-generated MR sets, maps real faults to 8 of 10 MR families, detects 7/20 equivariant-ML mutations versus 2/20 for a large-language-model (LLM)-prompted set and 0/20 for a literature set, and identifies reactor-physics counterexamples plus five obstructions that bound the current Translate signature. The evidence characterises NOETHER as an auditable, bounded MR-identification framework, not as a general-purpose or average-superiority claim.

CCS Concepts: • **Software and its engineering** → **Software testing and debugging**; *Software verification and validation*; • **Theory of computation** → *Algebraic semantics*.

Additional Key Words and Phrases: metamorphic testing, metamorphic relation identification, MetaPattern, operator algebra, program-family testability, test oracle problem, software testing foundations

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1 Introduction

Noether's first theorem is useful here only as a methodological analogy: a catalogue of observed invariants can sometimes be replaced by a derivation grounded in structure.¹ The replacement does not eliminate empirical work; it lifts the empirical step from per-instance enumeration to per-domain structural identification, one cycle of induction per stable domain rather than per program.

Software testing faces a related problem. Metamorphic Testing (MT), introduced by Chen et al. in 1998 [10], was admitted into the IEEE/ISO/IEC software-testing standards in 2022 [25] and is now

¹The theorem appeared in Noether's 1918 paper *Invariante Variationsprobleme*, addressing a question Hilbert and Klein had raised about energy conservation in general relativity. We do not invoke it as a theorem about programs: program semantics and metamorphic relations do not provide an action functional. The analogy is methodological only.

used for testing artificial intelligence (AI) and machine-learning (ML) systems [33, 47]. Its central artefact is the *metamorphic relation* (MR): a property that constrains how a program's outputs must covary across multiple executions, thereby substituting for an explicit oracle in domains where one cannot exist. Despite MT's maturity, MR identification, the task of deciding *which* properties hold for a program under test, remains its binding constraint. Practitioners report three recurring difficulties: high dependence on tester domain knowledge, mutually incompatible MR formulations for the same program, and low reuse of MR sets across teams or projects [33, 47]. Recent work has responded at the application layer, by mining MRs for particular domains, and at the integration layer, by automating search through evolutionary, mining-based, or large-language-model (LLM)-assisted pipelines [2, 48, 60]. The foundational layer has not advanced at the same pace.

The principal artefact at that foundational layer is the *MetaPattern* (MP): an equivalence class over MRs that captures a recurrent structural strategy a tester invokes when reasoning about program properties. MetaPatterns organise the otherwise unbounded MR design space into a small, interpretable scaffold; methods that exploit them, including the structured MR identification approaches METRIC [11] and METRIC+ [53], and the recent wave of LLM-prompted MR generators, depend on the scaffold's quality. Yet across the literature, MetaPattern catalogues continue to be assembled in the same way conservation laws were assembled before 1918: by induction over observed examples. A typical proposal lists k patterns drawn from cluster analysis or expert codification, demonstrates that the patterns "cover" some corpus of MRs to a target threshold, and stops. None of the existing MP proposals, including prior work on five reactor-physics patterns, answers the three questions that any structural account of MetaPatterns should answer:

- (1) **Origin.** *Why exactly these MetaPatterns and not others? What is the structural source of an MP, as distinct from an empirical regularity in the corpus on which it was induced?*
- (2) **Closure.** *Under what mathematical conditions is a discovered MP set *closed under a stated derivation operator*, in the sense that it is guaranteed not to miss patterns reachable through the operator from a structurally fixed input? Absolute completeness over all properties one might write over the underlying structure is a strictly stronger demand and remains open in general.*
- (3) **Transferability.** *When the program family changes (from reactor physics to natural-language inference, from numerical libraries to recommender systems), how does the MetaPattern set change, and can the new set be obtained without re-running the entire empirical induction in the new domain?*

We call this the *origin-closure-transferability gap*. The gap explains why MR sets continue to grow as one-off artefacts. Without a structural source, there is no clear boundary on what the pattern space contains. Without a transfer rule, a relation written for one program rarely moves cleanly to another, even when both programs share a deep structural template. These limitations follow from grounding MetaPatterns in observed examples rather than in the structure that makes the relations hold.

This paper proposes such a structural source for a deliberately bounded class of systems: operator-algebraic program families, i.e., equation-governed or relational-semantics families whose behaviour can be represented through operators, invariants, and equivalence-preserving transformations. We introduce **NOETHER**, a constructive framework that derives MetaPatterns from the operator-algebraic structure of the program family under test. The framework extracts MetaPatterns from invariants of an operator algebra $\mathcal{A}_P$ (formally defined in Section 3.1) that captures the program family's mathematical scaffolding. Given $\mathcal{A}_P$, NOETHER produces a MetaPattern set $\mathbb{M}(\mathcal{A}_P)$ (constructed by the algorithm of Section 3.2) together with a closure guarantee over the algebra-induced MR space. Given a different program family with a different algebra

$\mathcal{A}_P$, the same construction produces $\mathbb{M}(\mathcal{A}_P)$ without re-running empirical induction over observed MRs. The framework does not abolish induction: domain experts must still distil $\mathcal{A}_P$ from program semantics, and Section 6 states this limitation explicitly. What becomes algebraic is the downstream step from $\mathcal{A}_P$ to the MetaPattern set. Figure 1 summarises the architecture.

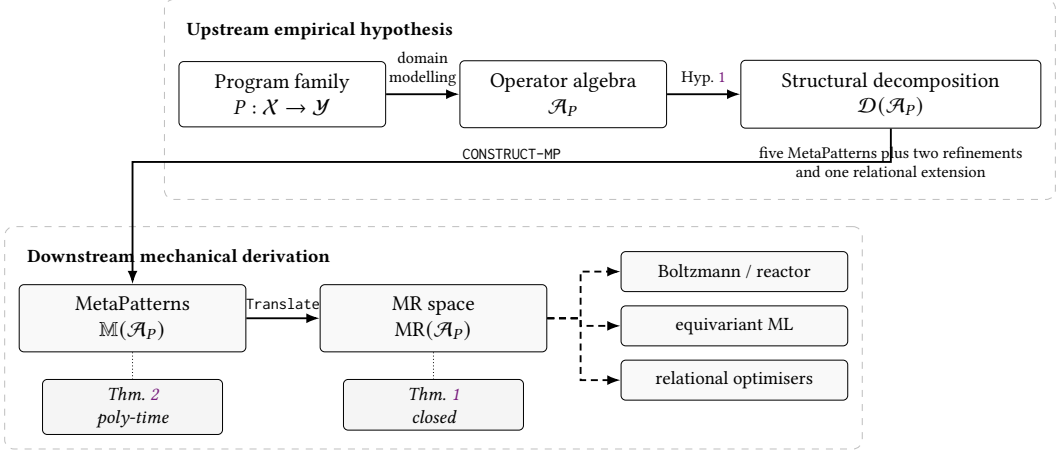


Fig. 1. NOETHER separates empirical algebra curation from mechanical MetaPattern derivation and MR instantiation.

We make five MR-identification contributions. The load-bearing theoretical result is an Invariance-Blindness Theorem (IBT): a falsifiable characterisation of which implementation faults an algebra-derived MR can and cannot detect, distinct from the intentionally modest by-construction closure guarantee.

- **C1 (framework).** We introduce NOETHER, a *layered* MR-identification framework for operator-algebraic program families: the upstream layer empirically curates a program-family operator algebra and structural decomposition (Hypothesis 1; partly distilled from a pressurized-water-reactor (PWR) MR corpus, §3.4.3), while the downstream CONSTRUCT-MP layer mechanically derives five MetaPatterns and ten MR families from that algebra.
- **C2 (theory).** We give a three-part theory for the algebra-induced MR space. Theorem 1 proves a *no-drop closure invariant*: every Translate-reachable MR over $\text{MR}(\mathcal{A}_P)$ is assigned to a canonical structural component and MetaPattern class. Theorem 2 gives polynomial-time constructibility under a finite generating set. The negative result refutes absolute completeness on $\mathcal{A}_{\text{PWR}}$ with two safety-analysis MRs and five pairwise-independent obstructions in Translate’s signature (§3.7, Appendix D); these obstructions are part of the contribution because they identify MR classes the current construction should not claim to reach. The Invariance-Blindness Theorem (§3.3) characterises exactly which implementation faults the G and T^* MetaPatterns can and cannot detect within the stated linear fault class.
- **C3 (canonical symbol system).** Instantiating NOETHER on a real program family maps a literature-derived PWR MR corpus onto one canonical vocabulary: five MetaPatterns, two family-layer refinements, one relational extension, and the ten MR-family symbols of Table 1. The same symbols are used throughout the theory, figures, evidence tables, and discussion, including the deflationary comparison to Murphy et al.’s six ML classes [39] (§5.5).

- **C4 (transferability).** We demonstrate *structural* transferability, not empirical superiority, by instantiating NOETHER on three structurally distinct algebras: Boltzmann reactor physics (§3.4), equivariant ML (§3.5), and relational query optimisers (§3.6), the last exercising the relational extension beyond the Lie-group / self-adjoint / time-reversal core.
- **C5 (evidence protocol).** We define an MR-identification evidence protocol around RQ1–RQ4: canonical coverage, origin/boundary explanation, cross-domain and real-fault recurrence, and secondary executability/blind-spot checks. The empirical evidence includes expert/search-derived MR corpora, 21 real scientific-computing faults mapped to 8 of 10 MR families, and an equivariant-ML mutation study in which the NOETHER-derived set detects 7/20 mutants versus 2/20 for an LLM-prompted set and 0/20 for a literature set. These results are used as executability, complementarity, and blind-spot evidence, not average-superiority evidence.

Scope and adoption value. This paper evaluates *MR identification* for operator-algebraic program families: whether a method can derive structurally justified MR classes from a program family’s governing equations or relational semantics. It does not evaluate MR *effectiveness* in the sense of average fault-detection rate, mutation score, or general defect-revealing superiority. The five Meta-Patterns (with two refinements and a relational extension) are curated by inspecting mathematical structures that recur across the program families we have studied; they are not derived from an algebraic axiom. The framework therefore has two layers: an *upstream layer* (curating $\mathcal{A}_p$ and its structural decomposition) that remains empirical and human, and a *downstream layer* (mechanically deriving $\mathbb{M}(\mathcal{A}_p)$ from $\mathcal{A}_p$) that is algorithmic and provable. NOETHER’s intended practical value is not to replace search-based MR generators in every setting, but to give high-structure domains an auditable source for MR families, a transferable symbol system, predictable blind spots, and a structural prior that can complement search, mining, or LLM-prompted generation.

Upstream modelling effort. NOETHER moves, rather than eliminates, one part of the identification burden. The upstream modelling task is to collect the governing equations or semantic rewrite rules, name the operators exposed by those equations, assign them to the current structural components, and record which program-interface controls and observations make the resulting family executable. The case studies report the downstream cost once $\mathcal{A}_p$ is available, but they do not establish a general human-effort distribution for algebra distillation; this remains a limitation and a measurement target for future work. The claim of this paper is therefore bounded: once a program-family algebra has been curated, CONSTRUCT-MP gives a closed, auditable derivation of the algebra-induced MR space and exposes the boundary of the current Translate signature.

The remainder of the paper follows a TOSEM-style method-paper structure. Section 2 reviews expert MR design, structured taxonomies, and search-based MR generation. Section 3 presents the NOETHER method, the two-level model from identified MR class to executable MR instance, the structural-component taxonomy, and the coverage rules. Section 4 defines RQ1–RQ4 and the evidence protocol. Section 5 reports canonical coverage, origin/boundary explanation, cross-domain and real-fault recurrence, and RQ4 executability/blind-spot checks. Section 6 discusses threats to validity and limitations. Section 7 states future work; Section 8 concludes.

2 Related Work

We organise prior work along four lines: MT/MR fundamentals and the long-standing identification bottleneck; structured MR identification through category, specification, and model scaffolds; automated MR identification through mining, search, learning, and LLM assistance; and MetaP-pattern catalogues, including a prior reactor-physics taxonomy. These lines differ in mechanism, but they share one limitation: their pattern structures are induced from observed MRs, mined

artefacts, or manually supplied models rather than derived from the mathematical structure that makes those MRs hold.

2.1 Metamorphic testing and the MR identification bottleneck

Metamorphic Testing was introduced by Chen, Cheung, and Yiu in 1998 to address the test-oracle problem [10], a general obstacle in software testing when expected outputs cannot be automatically determined [4]. An MR is a logical implication of the form $R_i(x_1, \dots, x_n) \Rightarrow R_o(P(x_1), \dots, P(x_n))$, where P is the program under test, R_i is an input relation, and R_o is an output relation. When R_i holds but R_o fails on actual executions, a fault is reported, without any need to know the absolute correct output of any single execution. Over two decades MT has become standard equipment for testing systems whose oracles are otherwise inaccessible: scientific computing, machine learning classifiers, autonomous vehicles, search engines, compilers, and large language models [33, 47].

The community has long acknowledged a single binding constraint on MT's effectiveness: MR identification. Surveys spanning twenty years agree that identifying high-quality MRs requires (i) deep familiarity with the program's functional semantics, (ii) substantive domain background (physical, mathematical, or linguistic, depending on the system), and (iii) the ability to convert that background into executable property assertions [33, 47]. Liu et al. [36] provide the canonical empirical evidence that even ad-hoc MRs detect a substantial fraction of mutants given modest tester training, while Murphy et al. [39] catalogued the structural properties of ML applications relevant to MR design and Xie et al. [59] demonstrated that MT scales effectively to supervised classifiers. Testers without sufficient domain background tend to identify only "trivial MRs". A 2024 survey explicitly identifies AI assistance, especially via large language models, as the most promising open avenue for closing the identification gap [33]; Saha and Kanewala [43] report that, on a benchmark of 709 mutants for supervised classifiers, current MR sets detect only 14.8%, which exposes the gap.

The bottleneck is also structural: even when MRs are identified, the field has accumulated little consensus on *how* they are identified. Different authors confront the same program and produce dissimilar MR sets. Once written, an MR rarely migrates: a relation drafted for one ML classifier seldom transports to another, and a relation drafted for one numerical solver seldom transports across solver families.

2.2 Structured MR identification

Structured MR identification has already moved the field away from ad hoc relation design. METRIC organises MR construction around an "input/output category" framework: the tester first identifies relevant categories of input transformations and output relations, then composes MRs from category pairs [11]. METRIC+ extends this scheme by enriching the category catalogue and providing systematic combination rules that reduce the human burden in category enumeration [53]. Other structured lines instantiate the scaffold through domain or artefact models: Segura et al. define MR patterns for query-based systems from recurrent query features and explicitly use relational algebra as the vocabulary for those patterns [46]; Li et al. construct MRs from tabular-expression specifications [32]; Yan et al. decompose the input side of scientific-computing MRs into basic input-pattern structures [61]; and Lin et al. organise scientific-software MRs hierarchically [35]. Together, these approaches remain the strongest existing attempt to give MR identification an explicit scaffold.

Our disagreement is not with this direction but with its grounding. The categories, patterns, tables, and input decompositions are introduced through expert curation, artefact-specific structure, or observed reuse, and are validated by empirical coverage on benchmark programs. This leaves

two of our three foundational questions unanswered. *Origin*: existing structured methods do not derive their categories from program-level operator algebra. Even query-based MR patterns use relational algebra as an organising vocabulary, not as a closed derivation system whose primitive structural components define an MR design space. *Closure*: these frameworks do not provide a mathematical condition under which the category set is complete; coverage is reported but not proved. The third question, *transferability*, is only partially addressed. The same category templates can be invoked across domains, but because the categories are not algebraically bound to specific program structures, transfer rests on an assumption of universality that remains open.

2.3 Automated MR identification

Automated MR identification attacks the bottleneck by searching for executable relations more directly. Early search and dynamic-inference work includes likely metamorphic-property inference from executions [52], search-based inference of polynomial MRs [64], and AutoMR’s discovery and cleansing of numerical MRs through search plus redundancy reduction [63]. MR-Scout mines MRs from existing test suites by extracting input-transformation and output-assertion patterns from test-case pairs and abstracting them into reusable relations [60]. GenMorph evolves MR candidates through genetic programming, co-evolving input transformations with output assertions and using mutation-killing as the fitness signal [2]. Shin et al. derive executable MRs from natural-language requirements through few-shot prompting of a large language model, with validation through an industrial questionnaire study with Siemens [48]; further LLM-assisted variants, including ChatGPT-driven MR generation [66], domain-customised GPTs for autonomous-driving simulators [65], and multi-agent retrieval-augmented pipelines for traffic-rule MRs [34], extend this template into safety-critical or rule-rich domains. Within ML-system testing, differential-testing approaches such as DeepXplore [42] use input-transformation invariance as a proxy oracle, complementary to the explicit MR formulation we adopt here.

A second strand uses program-structure, specification, documentation, or model features rather than test-case mining. Kanewala et al. [27] predict MRs of scientific software via graph-kernel learning over control-flow and data-dependency graphs, an early demonstration that program structure carries enough signal for automated MR discovery. MeMo extracts equivalence MRs from Javadoc comments [7]. Nolasco et al.’s MemoRIA [40] infers MRs by deriving an object-protocol abstraction and validating candidates through fuzzing plus SAT-based reduction, then evaluates on 22 Java subjects. Clark et al. generate MRs from causal graphs, making model misspecification an explicit boundary of the method [12]. Tao et al.’s Mettoc [54] applied MT to compiler testing using equivalence-preservation as the canonical relation. Ying et al. [62] formalise relationships between metamorphic-relation patterns into family trees, and Altamimi et al. [1] provide a recent systematic literature review of MR-automation work that catalogues the structural-vs-empirical-search divide our framework targets. Earlier LLM-assisted attempts include Zhang et al. [66] on ChatGPT-driven MR generation.

Each of these methods automates a different part of MR identification, but all treat the MR space as an empirical, artefact-derived, or model-supplied search space. MR-Scout’s recall is bounded by the relations already latent in existing tests. Search-based and evolutionary methods can discover useful numerical or executable MRs, but the reported cleansing and redundancy-reduction steps follow from the search representation rather than from an account of the mathematical origin of the surviving relations. GenMorph’s evolutionary search is shaped by the mutation-killing fitness function. LLM-prompted approaches are sensitive to prompt phrasing when no structural prior constrains generation. Documentation-, protocol-, and causal-model-based methods are more interpretable, but their boundary is the supplied artefact or model rather than a program-family

operator algebra. None of these methods can state, before search begins, which MR types lie outside its reach, because none has an algebraic account of the space being searched. The tradeoff is explicit: NOETHER requires a stronger upstream domain model and is less automatic at the start, but it supplies algebraic warrant, closure, and non-reachability information. It should therefore be read as a structural prior and audit layer for high-structure domains, not as a replacement for mining, search, or LLM prompting.

The intuition that symmetries furnish partial oracles predates this work. Gotlieb’s symmetric testing treats input permutations whose outputs must agree as a checkable oracle [20], and a follow-up study models its test quality on Java Card APIs [21]. Patel and Hierons survey such oracle strategies for non-testable systems [41], while Khritankov and Iakusheva systematically review metamorphic-relation derivation and group the recurring input transformations into six families [28]. In machine learning, related symmetry notions are studied empirically: Gruver et al. measure how much equivariance a trained network actually acquires [22], and Kaba et al. attain equivariance by learning canonicalization functions rather than constraining the architecture [26]. These lines name and exploit the symmetry-oracle intuition, yet each identifies relations per program, surveys them descriptively, or approximates symmetry inside a model. None derive metamorphic patterns algorithmically from the governing equations of a program family. NOETHER does not invent the symmetry layer; its contribution is a constructive derivation over the structural-component layer, together with closure and complexity results and a domain-transferable skeleton. In that setting, the symmetry oracle becomes a construction rule rather than a per-program insight.

2.4 MetaPattern catalogues and empirical adequacy

MetaPattern catalogues organise the post-hoc inventory of identified MRs into a smaller vocabulary of recurring strategies. Such catalogues typically result from clustering observed MRs, codifying expert intuitions about “kinds of properties testers reason about”, and validating the catalogue’s coverage on a benchmark MR corpus. Reactor-physics test suites have produced taxonomies of conservation, monotonicity, convergence, trajectory, and partial-order patterns [6, 31]; Zhou et al. [69] introduced an explicit *symmetry metamorphic relation pattern* as a reusable abstraction for deriving concrete MRs across applications; Lin et al. [35] proposed hierarchical MRs for scientific software; Ying et al. [62] formalises the relationships between MR patterns into *family trees* with explicit refinement and specialisation edges, and reports a state-of-the-art (SOTA) pattern hierarchy that organises previously proposed patterns into a single classification structure. The most recent state-of-the-art survey of MR generation [33] maps the field’s twenty-year output across pattern catalogues, mining, evolutionary search, and LLM-prompted methods. Adequacy frameworks such as the Pattern–Matrix Coverage Metric, and recent MT adequacy work based on MRs and source inputs [16], assess how thoroughly a given test suite or MR set occupies an already-defined testing space. They do not define coverage over the structural components that give rise to MR classes.

Ying et al.’s family-tree formalism [62] is the closest published cousin: both organise MR patterns into a hierarchy, but Ying et al.’s family tree is a refinement/specialisation tree rooted in informally-named categories (symmetry, additive, multiplicative), whereas NOETHER’s MetaPatterns are quotients of the algebra-induced MR space $\text{MR}(\mathcal{A}_p)$ under structural equivalence (Definition 5), each derived from a specific structural component. A family-tree node maps to one or more NOETHER MetaPatterns when it admits an operator-algebraic specification (Ying et al.’s “symmetry” node splits into m_{inv} and m_{adj}), while $\mathcal{B}_{\text{rel}}^*$ has no family-tree counterpart; the two are complementary, Ying et al. cataloguing inductively at the MR-instance level and NOETHER providing the algebraic warrant per pattern.

Such catalogues are useful, but each is, by construction, an empirical artefact. None of the existing catalogues can answer the three foundational questions of Section 1.

2.5 Convergent diagnosis

Across all four lines of work, two problems recur. *Unbounded MR emergence*: every new domain or new program produces fresh MR formulations. *Poor MR reusability*: identified MRs do not transport readily across programs that share deep structural similarities. Both problems follow from grounding MetaPatterns inductively. NOETHER replaces that inductive grounding with operator-algebraic grounding.

Relation to a separate line of work on minimum complete MR subsets. A separate line of work addresses the *orthogonal* problem of *selecting* a minimum complete subset from a *given* metamorphic-relation pool: taking an MR set as input, it characterises the smallest subset that preserves the pool's fault-detection power under a fixed fault model, proves the corresponding decision problem NP-hard, and provides approximation and exact (ILP) solvers. That work takes the MR set as given and optimises its cardinality; the present paper is concerned with the upstream question of where MRs and MetaPatterns *come from*: their constructive derivation from a program-induced operator algebra and algebraic closure under Translate. The two are complementary: NOETHER *generates* the algebra-induced MR space that a subset-selection procedure would then *minimise*. No theorem or empirical claim is shared: the present paper makes no minimality or subset-selection claim, and the subset-selection work makes no derivation or closure claim.

3 Proposed Method

This section presents NOETHER as an MR-identification method. The method takes a program-family governing equation/operator system as input, distils a program-induced operator algebra $\mathcal{A}_p$, decomposes the algebra into the five MetaPatterns, two family-layer refinements, and a relational extension, derives MetaPatterns and MR families from component invariants, and finally checks whether a concrete program interface exposes the controls and observations needed to instantiate an executable MR. The primary output is an *identified MR family*; an executable MR instance is a downstream realization that may be blocked by missing interface affordances.

Counting rule used throughout the paper. Expert coverage is binary: an expert MR set covers a structural component if at least one expert MR belongs to that structural component. NOETHER coverage is also binary: NOETHER covers a structural component if the method explicitly derives at least one MR class from the program-family operator algebra in that structural component. Equation structure alone is not counted as coverage unless a corresponding MR class is written down. This rule is used before any empirical result is interpreted.

Status of formal objects. Definitions in this section fix the objects over which the claims are made: the program-induced operator algebra, structural component, MetaPattern, MR family, Translate, algebra-induced MR space, and MR mode. Hypothesis 1 is empirical and refutable; it is not a theorem. Theorem 1, Theorem 2, and Theorem 3 are conditional results over those objects and the stated fault/signature classes. Theorem 1' is the deliberately stronger absolute-completeness conjecture, and Section 3.7 refutes it.

Running example. For the equivariant-ML instance, the family algebra $\mathcal{A}_{\text{equi}}$ contains an $E(3)$ group action on point-cloud inputs and an output representation on logits. CONSTRUCT-MP assigns this operator to G , derives the MetaPattern m_{inv} , instantiates the MR family $f_{\text{inv.eqv}}$, and produces an executable rotation MR ρ_{rot} when the interface exposes point coordinates and class logits.

The same chain is used throughout: operator in $\mathcal{A}_P \rightarrow$ structural component $\rightarrow$ MetaPattern $\rightarrow$ MR family $\rightarrow$ executable MR instance if the interface permits.

Operator-algebraic preliminaries and the structural decomposition appear in §3.1; the algebra-induced metamorphic relation space, the CONSTRUCT-MP construction, the algebraic closure theorem (Theorem 1), and the polynomial-time constructibility theorem (Theorem 2) in §3.2. The framework is then instantiated on three structurally distinct operator algebras: Boltzmann reactor physics in §3.4, equivariant machine learning in §3.5, and relational query optimisers in §3.6. The section concludes by falsifying the strictly stronger absolute-completeness conjecture, Theorem 1', on a fourth program family (PWR core diffusion) via two pairwise-independent counterexamples (§3.7).

3.1 Operator-algebraic preliminaries

This section introduces the algebraic apparatus used by NOETHER. We assume basic familiarity with group actions, equivalence classes, and quotient sets, but not with functional analysis. Each construct is defined at the level needed for the framework and later instantiated in reactor physics and equivariant machine learning.

We treat a program P as a (possibly partial) computable function $P : X \rightarrow \mathcal{Y}$, and a *program-induced operator algebra* $\mathcal{A}_{\mathcal{F}} = (O, \circ, \sim_{\mathcal{F}})$ collects the operators acting on X , $\mathcal{Y}$, or both, under a composition $\circ$ and a program-family equivalence $\sim_{\mathcal{F}}$. The decomposition below draws on eight structural-operator types, each defined formally in Appendix B: *symmetry groups* ($G: P(g \cdot x) = \rho(g) P(x)$), *order operators* ($O_{\leq}$: monotonicity and linearity), *self-adjoint operators* ($T^*: \langle Lx, y \rangle = \langle x, Ly \rangle$), *time-reversal operators* ($\mathcal{T}_{\text{rev}}^*$: anti-unitary involutions of a time coordinate), *limit operators* ($\mathcal{L}^*$: parametrised convergence $\mathcal{L}_{\theta} \rightarrow \mathcal{L}_*$), the *qualitative-dynamics* ($\mathcal{D}^*$) and *method-comparison* ($\mathcal{E}^*$) refinements, and the relational-equivalence extension ($\mathcal{B}_{\text{rel}}^*$, Definition 1).

3.1.1 Decomposition of an operator algebra. Given $\mathcal{A}_P = (O, \circ, \sim_{\mathcal{F}})$, we decompose O into the five MetaPatterns, two refinements, and a relational extension introduced above:

$$\mathcal{D}(\mathcal{A}_P) = (G, O_{\leq}, T^*, \mathcal{T}_{\text{rev}}^*, \mathcal{L}^*, \mathcal{D}^*, \mathcal{E}^*, \mathcal{B}_{\text{rel}}^*),$$

where G collects symmetry subgroups, $O_{\leq}$ monotone and linear operators, T^* self-adjoint operators, $\mathcal{T}_{\text{rev}}^*$ time-reversal operators, $\mathcal{L}^*$ limit operators, $\mathcal{D}^*$ qualitative-dynamics operators, $\mathcal{E}^*$ method-comparison operators, and $\mathcal{B}_{\text{rel}}^*$ relational-equivalence operators. We call these eight units the *structural components* of $\mathcal{D}(\mathcal{A}_P)$; the first five ($G, O_{\leq}, T^*, \mathcal{T}_{\text{rev}}^*, \mathcal{L}^*$) are the MetaPatterns, and the remaining three are the two refinements ($\mathcal{D}^*, \mathcal{E}^*$) and the relational extension ($\mathcal{B}_{\text{rel}}^*$).

Conservation laws are not an additional MetaPattern: by the Noether-style correspondence between symmetries and conserved quantities, a conservation/invariance relation is the G MetaPattern m_{inv} . When evaluation tables record conservation separately, they record the MR family $f_{\text{inv.con}}$ under G , not a sixth MetaPattern.

DEFINITION 1 (RELATIONAL-EQUIVALENCE EXTENSION). A relational-equivalence operator is a binary relation $\equiv_{\mathcal{R}}$ on expressions over an idempotent semiring $(S, \oplus, \otimes, \mathbf{0}, \mathbf{1})$ such that $E \equiv_{\mathcal{R}} E'$ iff E and E' are equal under all valid evaluation contexts of S , generated by a finite set $\mathcal{R}_{\text{rel}} = \{\mathcal{R}_1, \dots, \mathcal{R}_K\}$ of identity-preserving rewriting rules, each $\mathcal{R}_i$ an ordered pair $(\text{lhs}_i, \text{rhs}_i)$ of semiring expressions with the property $\text{eval}(\text{lhs}_i, D) =_{\text{bag}} \text{eval}(\text{rhs}_i, D)$ on every valid input D . The relational extension $\mathcal{B}_{\text{rel}}^*$ collects all such operators. It is empty for program families without idempotent-semiring rewriting structure (Boltzmann reactor physics, equivariant ML) and non-empty for those with it (relational query optimisers, §3.6).

Necessity and empirical status. We do not claim the eight structural components are necessary in an absolute sense or exhaustive; they are an *empirical curation* of mathematical structures that

recur across the program families we have studied, currently sufficient for these families but not provably necessary in general. This is NOETHER’s main upstream limitation. Induction is not eliminated but moved one level up, from “what MetaPatterns recur in observed MRs?” to “what algebraic structures recur across program families?”, after which the downstream derivation of $\mathbb{M}(\mathcal{A}_P)$ is mechanical and provably closed (Theorem 1). We state this sufficiency as an explicit empirical hypothesis with a documented out-of-scope catalogue.

HYPOTHESIS 1 (DECOMPOSITION SUFFICIENCY). *Let P be a program belonging to a program family $\mathcal{F}$ whose semantics admit an operator-algebraic representation $\mathcal{A}_P$ over (i) finite or finite-dimensional symmetry groups, (ii) partial orders with monotone or linear operators, (iii) self-adjoint operators with respect to a fixed inner product, (iv) anti-unitary involutions of a time coordinate, (v) limit operations of analytical (mesh-refinement, parameter-perturbation, or asymptotic) type, (vi) qualitative-dynamics attractors of an underlying ordinary-differential-equation / partial-differential-equation (ODE/PDE) system, (vii) inter-method comparison partial orders, and (viii) identity-preserving rewriting rule sets on an idempotent semiring. For such families the structural decomposition*

$$\mathcal{D}(\mathcal{A}_P) = G \dot{\cup} O_{\leq} \dot{\cup} T^* \dot{\cup} \mathcal{T}_{\text{rev}}^* \dot{\cup} \mathcal{L}^* \dot{\cup} \mathcal{D}^* \dot{\cup} \mathcal{E}^* \dot{\cup} \mathcal{B}_{\text{rel}}^*$$

is sufficient: every operator in $\mathcal{A}_P$ relevant to MR derivation is assigned to at least one MetaPattern, refinement, or the relational extension. Hypothesis 1 is an empirical hypothesis open to refutation.

REMARK 1 (SINGLE-CLASS INSTANCES). *For the three instantiated algebras we adopt a canonical π -template per non-empty structural component under which $|I_s/\sim_s| = 1$ (e.g. Boltzmann’s G MetaPattern is a single $\sim_s$ class covering both geometric quarter-rotations and energy-group permutations), so $|\mathbb{M}(\mathcal{A}_P)|$ equals the number of non-empty structural components per instance; a finer treatment (Step 3 of CONSTRUCT-MP) would split the corresponding MetaPattern.*

REMARK 2 (DECOMPOSITION SUFFICIENCY VS. Translate SUFFICIENCY). *Hypothesis 1 asserts the structural components suffice to assign every relevant operator, not that every MR over $\mathcal{A}_P$ is Translate-reachable from a single component invariant. The negative PWR result (§3.7) exhibits two MRs whose operators are assigned to components yet whose content is not in $\text{MR}(\mathcal{A}_{\text{PWR}})$, so Translate’s expressive form is a second, independent constraint beyond the decomposition.*

REMARK 3 (KNOWN AND CONJECTURED OUT-OF-SCOPE PROGRAM FAMILIES). *The hypothesis is open to refutation. Six families are known or conjectured to fall outside its image and to require an additional MetaPattern: (i) symplectic systems; (ii) sheaf-theoretic / categorical constructions; (iii) probabilistic / martingale invariants; (iv) topological invariants; (v) label-consistency for supervised learning (empirically witnessed by the §5.2 cat-(i) wrong-sign 0/5); and (vi) empirical parameter-distribution divergence (witnessed by the §5.2.1 pilot’s undetected cat-v-02/04/05). Each signals a candidate additional MetaPattern, to be admitted when the program family demands it; one (metric-stability M_{lip}) has an explicit Translate-template design (Remark 5), and Appendix D catalogues concrete out-of-scope MRs.*

REMARK 4 (DOMAIN-LEVEL OUT-OF-SCOPE). *In addition to the candidate-additional-MetaPattern families of Remark 3, four program-family classes are out of scope for NOETHER in its present form because they admit no operator-algebraic representation at all: general web applications, reinforcement-learning-from-human-feedback (RLHF) reward models / LLM agents, distributed-consensus protocols, and compiler-internal optimisations. NOETHER’s reach is therefore consistent with its scope precondition: systems under test (SUTs) from textbook-codified mathematical, physical, or relational-algebra domains are within reach, those from general software-engineering domains without such codification are not.*

REMARK 5 (METRIC-STABILITY CANDIDATE ADDITIONAL METAPATTERN, M_{lip}). *The most concrete candidate additional MetaPattern (Remark 3) is metric-stability: M_{lip} is the family of K -Lipschitz maps $(X, d_X) \rightarrow (\mathcal{Y}, d_Y)$, with induced MetaPattern m_{lip} the class of pointwise stability inequalities $d_Y(P(x'), P(x)) \leq K d_X(x', x)$ and a Translate template that perturbs $x' = x + \varepsilon u$ ($\|u\| = 1, |\varepsilon| < \delta$) and predicates a K -Lipschitz output bound. It would sit after $\mathcal{B}_{\text{rel}}^*$ in the canonical ordering (metric structure is independent of the eight existing component invariants), and Theorem 1's proof transfers since m_{lip} is single-MetaPattern. We do not commit to it as canonical here; it is the empirical orphan of the §3.4.3 audit, recorded in Appendix D.*

3.1.2 *The two-layer model: five MetaPatterns and ten MR families, and MR mode.* The model has two layers. Layer 1 (L1) is the set of MetaPatterns: five canonical structures on the solution space, each the program-level image of one algebraic structure (a group action, a partial order, a self-adjoint operator, a time-reversal involution, and a parametrised limit). Layer 2 (L2) is the set of MR families: the ten executable relations a test engineer enumerates, each instantiating one MetaPattern. Making the two layers explicit controls both how the decomposition is instantiated as executable tests and how detection independence is read.

Layer 1: the five MetaPatterns. The MetaPattern set is $\{G, O_{\leq}, T^*, \mathcal{T}_{\text{rev}}^*, \mathcal{L}^*\}$ (symmetry, order, self-adjointness, time reversal, and limits). Formally, each MetaPattern is the equivalence class of algebra-induced MRs that CONSTRUCT-MP derives from one such structure, named by the invariant it preserves: m_{inv} (Invariance) from G , m_{mono} (Monotonicity) from $O_{\leq}$, m_{adj} (Adjoint) from T^* , m_{rev} (Reversal) from $\mathcal{T}_{\text{rev}}^*$, and m_{conv} (Convergence) from $\mathcal{L}^*$. (The genus name *Adjoint* for m_{adj} is kept distinct from the child MR family $f_{\text{adj}, \text{self}}$.) Two further structures are refinements of these MetaPatterns rather than additional MetaPatterns, and they enter the model at Layer 2 as MR families. Qualitative dynamics $\mathcal{D}^*$ refines $O_{\leq}$: bounded-variation, no-overshoot, and Sturm-type oscillation invariants are order relations on a trajectory rather than on pointwise values (sharpened by T^* where the operator is Sturm–Liouville). Method comparison $\mathcal{E}^*$ refines $\mathcal{L}^*$: an accuracy partial order $M_1 \preceq M_2$ is an order on a family of approximants converging to the same limit. Representation-invariance under $\mathcal{L}^*$ specialises, in the exact idempotent-semiring case, to relational equivalence $\mathcal{B}_{\text{rel}}^*$, which lies outside the Lie-group / self-adjoint / time-reversal core. Conservation remains the Noether image of G (the m_{inv} entry of Table 2), not a separate MetaPattern. This two-layer reading is a presentation of the same decomposition, not a new hypothesis: the sufficiency claim (Hypothesis 1) and the closure and decidability results (Theorems 1, 2) are unchanged.

Layer 2: the ten MR families. An MR family is the set of executable MRs obtained by instantiating one MetaPattern from $\mathbb{M}(\mathcal{A}_P)$ under a fixed π -template and a chosen MR mode (Definition 2); it is the unit a test engineer enumerates and one row of the empirical coverage matrix (§5.2.1). Table 1 lists the ten families used in this paper. Each family is written $f_{\text{parent}, \text{child}}$: a sans-serif family glyph whose subscript names its parent MetaPattern (before the dot) and the instantiating mechanism (after), so the parent is recoverable from the symbol; this glyph is tier-distinct from the MetaPattern symbol $m_{\bullet}$ and the executable-MR symbol $\rho_{\bullet}$. The map from the five MetaPatterns to the ten families is one-to-many and program-family-dependent (Figure 2); which families are non-empty is determined by $\mathcal{A}_P$. A single MetaPattern may produce several families (the G MetaPattern m_{inv} produces both the equivariance family and, by the Noether correspondence of §3.1.1, the conservation family), consistent with Step 3 of CONSTRUCT-MP and Remark 1. The chain is therefore the two layers: MetaPatterns (L1) $\rightarrow$ MR families (L2); the family layer refines the MetaPattern set rather than introducing a parallel taxonomy.

Table 1. Ten MR families instantiate the five MetaPatterns under tier-distinct family symbols.

Family	MetaPattern	Mode	Invariant
$f_{\text{inv.eqv}}$ equivariance	G	I	$P(\pi(g)x) = \rho(g) P(x)$
$f_{\text{inv.con}}$ conservation	G	I	$Q(P(x)) = Q(x)$ (Noether charge Q)
$f_{\text{adj.self}}$ self-adjoint	T^*	I/M	$A=A^\top \Rightarrow$ structure-invariant; real spectrum
$f_{\text{adj.dual}}$ adjoint duality	T^*	M	$\langle L^{-1}s, q \rangle = \langle s, (L^*)^{-1}q \rangle$
$f_{\text{rev.traj}}$ trajectory reversal	$\mathcal{T}_{\text{rev}}^*$	I	$\Phi_t(\Theta \Phi_t x) = \Theta x$ (round trip)
$f_{\text{mono.stat}}$ static order	$O_{\leq}$	I	$u \leq v \Rightarrow Pu \leq Pv;$ $x \geq 0 \Rightarrow Px \geq 0$
$f_{\text{mono.shape}}$ dynamic shape	$O_{\leq} (\mathcal{D}^* \text{ refinement})$	I	$Z(Px) \leq Z(x)$ (no spurious extrema)
$f_{\text{conv.lim}}$ convergence	$\mathcal{L}^*$	I	$P_{h_2}x \rightarrow P_{h_1}x$; fixed-point self-consistency
$f_{\text{conv.rate}}$ accuracy-order	$\mathcal{L}^* (\mathcal{E}^* \text{ refinement})$	M	$\ \text{err}(M_2)\ \leq \ \text{err}(M_1)\ $
$f_{\text{conv.repr}}$ representation-invariance	$\mathcal{L}^* (\mathcal{B}_{\text{rel}}^* \text{ extension})$	M	$P^{(r_1)}x = P^{(r_2)}x$

MR mode: input-orbit versus implementation-orbit. An MR tests a compatibility law by varying one of two axes while holding the other fixed.

DEFINITION 2 (MR MODE). A Mode-I (input-orbit) MR fixes a single implementation P and varies the input along the MetaPattern structure: $R(P; x, t(x))$ with t a group action, a mesh refinement, or a time reversal. A Mode-M (implementation-orbit) MR fixes a single input and varies the implementation among realisations $P^{(1)}, \dots, P^{(k)}$ of the same ideal operator (different methods, drivers, storage layouts, or parallelisations): $R(P^{(1)}, \dots, P^{(k)}; x)$. A Mode-M MR is a relative oracle: a violation localises the fault to the disagreeing set rather than to a single realisation, so localisation needs an additional discriminator (a third method, a control input on which the realisations must agree, or pre/post version closure).

Fault classes and detection independence. Each family R_i tests one compatibility law and, by the Invariance-Blindness reading (Theorem 3 for G and T^*), detects exactly the faults that break it. R_i independently distinguishes its fault class when some fault is detected by R_i and no other family (a unique-detection witness). Each family of Table 1 carries such a witness, including at least one non-crash fault, in the corpus of §5.2.1. This is witness-level necessity, not pairwise orthogonality (families sharing a MetaPattern may co-fire), and the minimal detecting subset is deferred to future work (§7).

3.2 Deriving MR classes with CONSTRUCT-MP

3.2.1 Algebra-induced metamorphic relations. Before defining algebra-induced MRs we make explicit the Translate operator that converts a single component invariant into an executable MR. Without this definition, the closure result of Theorem 1 would quantify over an undefined construct.

DEFINITION 3 (COMPONENT INVARIANT). Let $s \in \mathcal{D}(\mathcal{A}_P)$ be one of the eight structural components. A component invariant of $\mathcal{A}_P$ under s is a pair $\iota = (\Phi, \pi)$ where Φ is a finite set of operators drawn from s , and π is a relation $\pi \subseteq (\mathcal{X} \times \mathcal{Y})^k$ for some arity $k \geq 1$ such that for every $P \in \mathcal{F}$ and every choice of operators $\phi_1, \dots, \phi_n \in \Phi$, the tuple $((x_i, P(x_i)))_{i=1}^k$ obtained by applying $\phi_1, \dots, \phi_n$ to

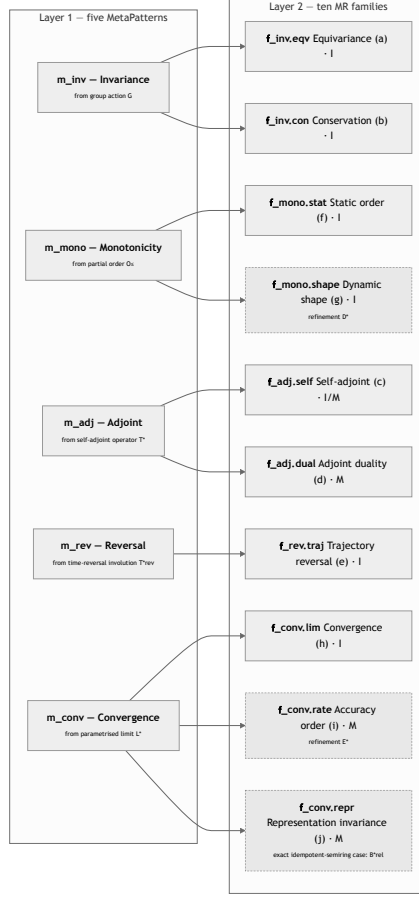


Fig. 2. The two-layer map links five MetaPatterns to ten MR families, with refinements and the relational extension entering at the family layer.

a base input $x \in \mathcal{X}$ in the canonical order specified by s satisfies π . We write $\mathcal{I}_s$ for the set of component invariants under s and $\sim_s$ for the equivalence relation $\iota \sim_s \iota'$ iff $\Phi = \Phi'$ and π, π' define the same constraint up to relabelling of input/output coordinates.

The relation $\sim_s$ is reflexive, symmetric, and transitive by construction (set equality and constraint equality are equivalence relations); hence $\sim_s$ is a genuine equivalence relation on $\mathcal{I}_s$, justifying the quotient in Step 3 of CONSTRUCT-MP below.

DEFINITION 4 (Translate). The Translate operator is the function

$$\text{Translate} : \mathcal{I}_s \times \{s\} \longrightarrow \text{MR}(P)$$

that maps a component invariant $\iota = (\Phi, \pi)$ under structural component s to the metamorphic relation

$$\rho_{\iota,s} : \forall x \in \mathcal{X}, \forall \phi_1, \dots, \phi_n \in \Phi : \pi((x_i, P(x_i))_{i=1}^k) \text{ holds,}$$

where the input tuple $(x_i)_{i=1}^k$ is generated by applying the ϕ_j in the canonical order specified by s (the convention is fixed per structural component: e.g., for $s = G$, $x_i = g_i \cdot x_0$ with g_i enumerated by

group orbit; for $s = T^*$, the tuple ranges over the two arguments of the inner product). Translate is well-defined because (i) the per-structural-component canonical order resolves any ambiguity in tuple generation, and (ii) $\sim_s$ -equivalent invariants are mapped to MRs that are $\sim$ -equivalent as logical statements over P . Per-structural-component instantiations of Translate are tabulated in Appendix D.

DEFINITION 5 (ALGEBRA-INDUCED MR). Let $\mathcal{A}_P$ be a program-induced operator algebra and let ρ be an MR over a program P . We say ρ is induced by $\mathcal{A}_P$, written $\rho \in \text{MR}(\mathcal{A}_P)$, when there exist (i) a structural component $s \in \mathcal{D}(\mathcal{A}_P)$, (ii) a component invariant $\iota \in \mathcal{I}_s$, and (iii) the equality $\rho = \text{Translate}(\iota, s)$ holds in the sense of Definition 4. Properties of P that constrain executions but cannot be obtained as $\text{Translate}(\iota, s)$ for any single structural component s and any invariant $\iota \in \mathcal{I}_s$ are not algebra-induced in this sense; concrete examples are catalogued in Appendix D.

3.2.2 Construction of the MetaPattern set. We present the deductive procedure CONSTRUCT-MP that maps an algebra-decomposition to a MetaPattern set.

Step 1: Invariant extraction. For each structural component s in $\mathcal{D}(\mathcal{A}_P)$, compute the set of invariants $\mathcal{I}_s$ of $\mathcal{A}_P$ under s .

Step 2: MR derivation. For each invariant $\iota \in \mathcal{I}_s$, derive the MR family $\mathcal{R}(\iota) = \{ \rho \in \text{MR}(\mathcal{A}_P) \mid \rho = \text{Translate}(\iota', s), \iota' \sim_s \iota \}$.

Step 3: Quotient (per equivalence class). For each $\sim_s$ -equivalence class $[\iota] \in \mathcal{I}_s / \sim_s$, form the MetaPattern $m_{s, [\iota]} = \mathcal{R}(\iota)$.

Step 4: Aggregation. Return $\mathbb{M}(\mathcal{A}_P) = \bigcup_{s \in \mathcal{D}(\mathcal{A}_P)} \{ m_{s, [\iota]} : [\iota] \in \mathcal{I}_s / \sim_s \}$, so $|\mathbb{M}(\mathcal{A}_P)| = \sum_s |\mathcal{I}_s / \sim_s| \geq$ the number of non-empty structural components.

The four steps are intentionally stated as a textual algorithm; the supplementary package includes the corresponding implementation sketch.

REMARK 6 (METAPATTERN GRANULARITY). Each MetaPattern $m_{s, [\iota]}$ corresponds to a single $\sim_s$ -equivalence class of invariants in structural component s . A structural component carrying $k_s > 1$ such classes (for instance geometric quarter-rotations and energy-group permutations within G on $\mathcal{A}_{\text{Boltz}}$, when treated as separate $\sim_s$ classes) contributes k_s distinct MetaPatterns. In the three instances of §§3.4–3.6 we group invariants within a structural component under a single $\sim_s$ -equivalence class (a single canonical π -template per non-empty structural component; see Remark 1), so $|\mathbb{M}(\mathcal{A}_P)|$ equals the number of non-empty structural components in those instances. The Invariance-Blindness analysis (§3.3) applies directly to each $m_{s, [\iota]}$ at its native per-class granularity.

3.2.3 Algebraic closure under Translate, the canonical MetaPattern ordering, and out-of-scope MRs.

DEFINITION 6 (CANONICAL METAPATTERN ORDERING). We adopt the strict total order on structural components

$$G > O_{\leq} > T^* > \mathcal{T}_{\text{rev}}^* > \mathcal{L}^* > \mathcal{D}^* > \mathcal{E}^* > \mathcal{B}_{\text{rel}}^*.$$

An MR derivable through multiple structural components is assigned to the highest-priority structural component in this order. The placement of $\mathcal{B}_{\text{rel}}^*$ at the bottom reflects its semiring-rewriting nature, which sits algebraically downstream of the non-relational structural components: rewriting equivalences typically depend on the program family's input-perturbation, order, and method-comparison structure rather than the converse.

THEOREM 1 (ALGEBRAIC CLOSURE UNDER Translate). Let $\mathcal{A}_P$ be a program-induced operator algebra with decomposition $\mathcal{D}(\mathcal{A}_P)$ as in Section 3.1.1, and let $\mathbb{M}(\mathcal{A}_P) = \text{CONSTRUCT-MP}(\mathcal{D}(\mathcal{A}_P))$. Then for every $\rho \in \text{MR}(\mathcal{A}_P)$ in the sense of Definition 5, there exists a unique $m \in \mathbb{M}(\mathcal{A}_P)$ such that $\rho \in m$, where uniqueness is determined under the canonical MetaPattern ordering.

REMARK 7 (SCOPE OF THEOREM 1). *Theorem 1 is a closure statement, not a completeness claim over arbitrary properties one might assert about P 's executions. It quantifies over $\text{MR}(\mathcal{A}_P)$ as defined by Definition 5, namely the algebra-induced MRs reachable through the Translate operator from a single component invariant. Three concrete classes of MRs lie outside this scope and are not covered by the theorem; they are documented in Appendix D:*

- (1) Probabilistic MRs without operator-algebraic representation: e.g. MRs that constrain output distributions (such as “the entropy of $f(\mathbf{x})$ should not decrease under input augmentation A ”) when the augmentation is not expressible as an operator in $\mathcal{O}$.
- (2) Input-distribution MRs not expressible over $\mathcal{A}_P$ operators: e.g. adversarial-perturbation MRs “ $\|\delta\|_p \leq \epsilon \Rightarrow f(\mathbf{x} + \delta) = f(\mathbf{x})$ ” whose perturbation set is not a group action on $\mathcal{X}$.
- (3) Compositional MRs spanning multiple structural components under non-Translate derivations: MRs requiring simultaneous use of two component invariants in a way that is not the canonical-MetaPattern-ordering reduction of Definition 6.

The strictly stronger statement that every MR formulable as a property over $\mathcal{A}_P$'s operators (without restricting to Translate-reachable derivations) is contained in some $m \in \mathbb{M}(\mathcal{A}_P)$ is identified as Theorem 1' in Appendix D. Section 3.7 and Appendix D establish that this stronger statement is false on the PWR core diffusion algebra $\mathcal{A}_{\text{PWR}}$, by exhibiting two concrete counterexamples (ρ_{nonadd} and $\rho_{\text{MTC-bor}}$, where MTC denotes moderator temperature coefficient) whose obstructions identify five structurally independent extensions of Translate's signature. The combined open problem (whether such an extended Translate preserves Theorem 1's closure and Theorem 2's polynomial-time constructibility) is the principal open question for follow-up work.

A sceptical reading might object that the by-construction status of Theorem 1 makes it near-tautological. We acknowledge that the closure result is by-construction within the explicit scope of Definition 5 (which fixes $\text{MR}(\mathcal{A}_P)$ as the Translate-image of $\mathcal{A}_P$). The substantive value lies in what the theorem then enables: Theorem 1 converts an empirical-adequacy claim (“our pattern grid covers $X\%$ of observed MRs”) into a structural-adequacy claim (“our pattern grid is exhaustive of the algebra-induced MR space, modulo the explicit out-of-scope classes of Remark 7”). Empirical adequacy frameworks built on inductive pattern grids do not guarantee algebraic closure even in this bounded sense; NOETHER guarantees it, and the obligation imposed on any user of the framework is then to verify that no Translate-reachable MR is dropped by CONSTRUCT-MP, a verification that is mechanical once the algebraic input has been fixed.

The theorem still imposes a checkable obligation on the framework. If an MR is induced by an invariant in $\mathcal{D}(\mathcal{A}_P)$ but cannot be assigned by CONSTRUCT-MP, then at least one component (invariant extraction, Translate, structural equivalence, or the canonical ordering) is underspecified. Thus the result is not an empirical coverage claim; it is a well-formedness and closure guarantee for the downstream construction once the algebraic input has been fixed. The framework's additional theoretical results complement Theorem 1: Theorem 2 bounds the construction's polynomial-time complexity under a finite generating set; Theorem 1' (Conjecture D), which makes the stronger claim of absolute completeness over arbitrary properties expressible in $\mathcal{A}_P$, is falsified on $\mathcal{A}_{\text{PWR}}$ (§3.7); the falsification identifies five structural obstructions in Translate's signature (Appendix D), with five further candidate dimensions surveyed on $\mathcal{A}_{\text{equi}}$ and $\mathcal{A}_{\text{rel}}$ (§3.6).

REMARK 8 (SCOPE FOR THE RELATIONAL EXTENSION $\mathcal{B}_{\text{rel}}^*$). *Theorem 1's closure extends to $\mathcal{B}_{\text{rel}}^*$ when the rewriting rule set $\mathcal{R}_{\text{rel}}$ is finite (e.g. the heuristic-rewrite normal forms of TPC-H-class queries [57, 68]); unbounded recursive rewriting (non-terminating Datalog) is out of scope (Remark 3).*

3.2.4 Complexity and decidability.

THEOREM 2 (COMPLEXITY OF CONSTRUCT-MP UNDER A FINITE GENERATING SET). *Suppose $\mathcal{A}_P$ admits a finite generating set $\text{gen}(\mathcal{A}_P)$ of cardinality n , with each generator's invariant computation assumed to terminate in time t_i . Then $\mathbb{M}(\mathcal{A}_P)$ is computable in time $O(n \cdot \max_i t_i \cdot \log n)$. This is a complexity bound on the construction rather than a decidability result: termination of per-structural-component invariant extraction is assumed within t_i , and for the relational extension $\mathcal{B}_{\text{rel}}^*$ general query equivalence is undecidable, so only the fixed-rule-set fragment is covered (Remark 9). The factors t_i absorb the realised per-generator invariant cost: for a finite symmetry group the relevant t_i can scale with the group order $|G|$, which may itself grow exponentially in the number of generators (for example $(\mathbb{Z}_2)^k$ has $|G| = 2^k$), so the bound is polynomial in n and the realised generator costs (output-sensitive) rather than necessarily polynomial in a compact group presentation.*

REMARK 9 (DECIDABILITY FOR $\mathcal{B}_{\text{rel}}^*$ UNDER A FIXED RULE SET). *The polynomial-time bound holds for $\mathcal{B}_{\text{rel}}^*$ under a fixed rule set $\mathcal{R}_{\text{rel}}$: although first-order/bag-semantics query equivalence is undecidable in general, equivalence under a fixed rule set is decidable and SMT-verifiable on substantial benchmark fractions [57, 68].*

Per-structural-component cost. The per-generator invariant-extraction cost is $O(|G|^2)$ for finite symmetry groups, $O(d_G^2)$ over the Lie-algebra basis for finite-dimensional Lie groups (e.g. $\text{SO}(3)$, $d_G = 3$), and polynomial in a user-supplied truncation K for finitely-generated infinite-discrete groups; the per-structural-component breakdown and the infinite-group truncation discussion are in supplementary S1. For a finite group $|G|$ may grow exponentially in the number of generators (e.g. $(\mathbb{Z}_2)^k$ has $|G| = 2^k$), so Theorem 2's bound is output-sensitive in the realised generator costs rather than polynomial in a compact group presentation.

3.2.5 The principal limitation. NOETHER replaces inductive grounding with algebraic grounding downstream of $\mathcal{A}_P$. Upstream, the distillation of $\mathcal{A}_P$ from a program family remains a human task. This limitation is central to the framework's scope: NOETHER does not automate domain modelling, but it turns the step after domain modelling into a well-defined construction. A second limitation, made explicit in §3.7 and Appendix D, is that even when $\mathcal{A}_P$ is fully specified, Translate's present signature (single-component, first-order π -template, single partial-order direction, operating on $P(x)$ tuples rather than on operator-spectrum quantities) systematically excludes a class of MRs that engineering practice treats as standard: non-additivity of operator-composition functionals and higher-order mixed parametric dependences. The principal limitation is therefore twofold: the upstream distillation of $\mathcal{A}_P$, and the present signature of Translate.

The downstream layer is mechanical: Theorem 1 (algebraic closure under Translate) and Theorem 2 (polynomial-time constructibility) operate on a given structural decomposition. The upstream layer remains empirical: the structural decomposition is Hypothesis 1, an open empirical hypothesis with six documented out-of-scope program-family classes (Remark 3). Theorem 1' (absolute completeness) is open. Sections 3.4, 3.5, and 3.6 instantiate NOETHER on three structurally distinct program families.

3.3 The Invariance-Blindness Theorem

Theorem 1 guarantees that CONSTRUCT-MP drops no Translate-reachable MR; it is, by construction, silent about what the derived MRs can detect. This subsection supplies that missing content: a limiting characterization of the faults an algebra-induced MR cannot detect. Throughout we work in the linear operator-implementation fault class (Definition 7); scope and assumptions are collected in Remark 12.

DEFINITION 7 (LINEAR PROGRAM AND FAULT CLASS). *A linear program is $P_L : \mathbb{R}^N \rightarrow \mathbb{R}^N$, $P_L(x) = Lx$, indexed by an operator $L \in \Theta := \mathbb{R}^{N \times N}$ ($\dim \Theta = N^2 < \infty$). The reference (correct) program is*

P_{L^*} ; an operator-implementation fault is $L = L^* + \Delta$ with $\Delta \in \Theta$. (For nonlinear solvers, Θ is the operator's finite parameter vector and E_s below its first-order linearization.)

DEFINITION 8 (STRUCTURAL DEFECT AND DETECTION KERNEL). *Let structural component $s \in \mathcal{D}(\mathcal{A}_P)$ carry a structure T_s (for $s = G$, a group action $A \in G \leq GL(\mathbb{R}^N)$; for $s = T^*$, the inner-product adjoint). The algebra-induced MR $\rho_{L,s} = \text{Translate}(\iota, s)$ asserts a structural predicate $\Pi_s[P_L]$ over a witness set $W = S_s \times X_0$ ($S_s \subseteq T_s$, $X_0 \subseteq \mathbb{R}^N$). For $s = G$ the per-witness defect is the commutator applied to an input,*

$$E_G(L; A, x) = P_L(Ax) - AP_L(x) = [L, A]x,$$

linear in L ; for $s = T^$, $E_{T^*}(L) = L - L^\top$. The detection kernel is the set of faults the MR fails to flag,*

$$\ker(\rho_{L,s}) := \{L \in \Theta : E_s(L; w) = 0 \ \forall w \in W\}.$$

If X_0 spans $\mathbb{R}^N$ then $\ker(\rho_{L,s}) = \bigcap_{A \in S_s} C(A)$ with $C(A) = \{L : [L, A] = 0\}$ the commutant of A .

DEFINITION 9 (STRUCTURE-PRESERVING FAULT). *L is T_s -compatible iff it satisfies the full structural identity that validates $\rho_{L,s} : E_s(L; w) = 0$ for all witnesses $w \in W_s^{\text{full}}$ (for $s = G$, $[L, A] = 0$ for all $A \in T_s$). Write $C(T_s)$ for the T_s -compatible set (the commutant of the whole group, i.e. the equivariant operators).*

DEFINITION 10 (FAITHFULNESS). *A finite witness set W_s^{test} is faithful iff $\ker(\rho_{L,s}) = C(T_s)$; equivalently, since E_s is linear in L , $\text{rank } J_{\text{test}} = \text{rank } J_{\text{full}}$, where J collects the rows $L \mapsto E_s(L; w)$.*

LEMMA 1 (REACHABILITY OF FAITHFULNESS). *If Θ is finite-dimensional and E_s is linear in L , then a finite faithful witness set W_s^{test} exists.*

PROOF. Each witness w contributes a linear functional $a_w : L \mapsto E_s(L; w) \in \Theta^*$. The span $V_{\text{full}} = \text{span}\{a_w : w \in W_s^{\text{full}}\} \subseteq \Theta^*$ is a subspace of the finite-dimensional dual, hence finite-dimensional; pick finitely many witnesses W_s^{test} whose functionals span V_{full} . As $W_s^{\text{test}} \subseteq W_s^{\text{full}}$ the intercepts agree, so the two systems have the same solution set, i.e. $\ker(\rho_{L,s}) = C(T_s)$. $\square$

Lemma 1 is the crux: tightness is *not* tautological but follows from finite fault-dimensionality, so a finite executable MR can pin the exact blind set. We now characterize that set.

THEOREM 3 (INVARIANCE-BLINDNESS, G AND T^* METAPATTERNS). *Let the fault class be linear (Definition 7), let E_s be linear in L (holds for $s \in \{G, T^*\}$), and let the test witnesses be faithful (Definition 10). Then*

$$\ker(\rho_{L,s}) = C(T_s) = \{L : P_L \text{ preserves } T_s\}.$$

That is, a linear operator-implementation fault survives the structural MR if and only if it preserves the structure T_s that the MR exploits.

PROOF. ($\supseteq$) If $L \in C(T_s)$ then $E_s(L; w) = 0$ on $W_s^{\text{full}} \supseteq W_s^{\text{test}}$, so $\rho_{L,s}$ passes and $L \in \ker$. ($\subseteq$) If $L \in \ker$ then $E_s(L; w) = 0$ on W_s^{test} ; by faithfulness (Definition 10) $\ker(\rho_{L,s}) = C(T_s)$, so $L \in C(T_s)$. $\square$

REMARK 10 (PER-METAPATTERN GRANULARITY). *Under Step 3 of CONSTRUCT-MP, each MetaPattern $m_{s,[\iota]}$ corresponds to a single $\sim_s$ -equivalence class of invariants; the per-invariant statement of Theorem 3 therefore applies directly to each $m_{s,[\iota]}$ without aggregation. When the per-instance grouping of Remark 1 treats a non-empty structural component s as a single $\sim_s$ class, $m_{s,[\iota]}$ coincides with the MetaPattern reported in the empirical sections.*

REMARK 11 (WORKED INSTANCE: ADVECTION-DIFFUSION, TRANSLATION SUB-CLASS). *On a periodic grid the constant-coefficient operator $c \cdot \nabla + \alpha \nabla^2$ commutes with every grid translation for all constants (c, α) ; hence $E_G(L; \text{shift}) \equiv 0$ on the whole constant-coefficient family, so $\ker(\rho_{L,G})$ contains it, including the advection-speed fault $c \mapsto c'$. The translation MR is therefore blind to advection-speed*

errors, matching the secondary executability sanity check discussed in §5. A spatially inhomogeneous coefficient $c(x)$ breaks translation equivariance, so $E_G \neq 0$ and the fault is detected. A faithfulness check (one generator plus an input basis is unisolvent) is reported in supplementary S10.

COROLLARY (SINGLE-METAPATTERN INCOMPLETENESS). For any structural component s with $C(T_s) \neq \{L^*\}$ (a nontrivial structure-preserving fault exists, noting $C(T_s) \supseteq \{L^*\}$ always), the single-MetaPattern battery $\{\rho_{i,s}\}_i$ is incomplete: its kernel contains that fault.

COROLLARY (COMPLETENESS REQUIRES A TRIVIAL JOINT KERNEL). A family of oracles $\{O_j\}$ detects every nontrivial fault only if $\bigcap_j \ker(O_j) = \{L^*\}$. Completeness therefore requires oracles whose structural kernels intersect trivially, not additional MRs of the same MetaPattern.

COROLLARY (DIFFERENTIAL ORACLE IS THE COMPLEMENTARY BLIND SPOT). The neutral cross-implementation differential oracle O_{diff} , which flags a fault when two independent implementations disagree beyond a discretisation-calibrated tolerance, has kernel $\ker(O_{\text{diff}}) = \{\text{common-mode faults (identical effect on both implementations)}\}$. Since common-mode and T_s -compatible are distinct conditions, $\ker(\rho_{i,s}) \cap \ker(O_{\text{diff}})$ is strictly smaller than either: the MR battery and the differential oracle are complementary, and a fault escapes both only if it is simultaneously structure-preserving and common-mode.

REMARK 12 (SCOPE AND ASSUMPTIONS). **(R1)** Theorem 3 is stated for the linear operator-implementation fault class (Definition 7); it does not extend to arbitrary nonlinear faults. **(R2)** Detection is exact-arithmetic; an executable MR with tolerance τ has kernel $\ker_\tau \supseteq \ker$, so faults below τ are false negatives. Theorem 3 is the $\tau \rightarrow 0$ limit; the finite- τ regime is outside the identification claim and is treated as a limitation in §6. **(R3)** Linearity of E_s holds for G and T^* (commutator; symmetric part). It fails for $O_\leq$ (monotonicity is an inequality, a cone not a subspace), for $\mathcal{T}_{\text{rev}}^*$ (reversibility $RLR^{-1} - L^{-1}$ involves the inverse), and for $\mathcal{L}^*$ (Richardson ratio of norms); for these MetaPatterns only the sufficient direction ($\supseteq$) holds, or a linearized subclass. **(R4)** The sufficient direction is by definition; the content is the $\subseteq$ direction together with Lemma 1 and the kernel characterization.

3.4 Boltzmann instantiation: from transport to diffusion to burnup

This section instantiates NOETHER on the Boltzmann transport equation and traces the construction through neutron transport, diffusion, and burnup.

3.4.1 The Boltzmann program family and its operator algebra. The Boltzmann transport equation governs the distribution of neutral particles. In its time-independent eigenvalue form for fission systems:

$$\begin{aligned} \hat{\Omega} \cdot \nabla \psi(\vec{r}, \hat{\Omega}, E) + \Sigma_t(\vec{r}, E) \psi \\ = \int \int \Sigma_s(\vec{r}, E' \rightarrow E, \hat{\Omega}' \rightarrow \hat{\Omega}) \psi dE' d\hat{\Omega}' + \frac{1}{k} \chi(E) \int v \Sigma_f(\vec{r}, E') \psi dE'. \end{aligned} \quad (1)$$

The program-induced operator algebra $\mathcal{A}_{\text{Boltz}}$ contains

$$\{\mathcal{G}_{\text{geom}}, \mathcal{R}_E, \mathcal{L}_\Sigma, \mathcal{L}_v, \mathcal{L}^*, \mathcal{T}, \mathcal{L}_h, \mathcal{D}_{\text{Bate}}, \mathcal{E}_{\text{cmp}}\}.$$

Decomposed along the eight structural components of Section 3.1.1, $\mathcal{A}_{\text{Boltz}}$ has entries in seven structural components (with $\mathcal{B}_{\text{rel}}^*$ empty under the absence of idempotent-semiring rewriting structure). Every operator listed has an unambiguous status in the established theory of neutral-particle transport [6, 31].

3.4.2 *Running CONSTRUCT-MP on $\mathcal{A}_{\text{Boltz}}$.* Steps 1–4 of CONSTRUCT-MP produce five MetaPatterns: m_{inv} (invariance/equivariance), m_{mono} (parameter-monotonicity), m_{adj} (self-adjoint duality / adjoint reciprocity), m_{rev} (time-reversal compatibility), and m_{conv} (discretisation convergence), together with two MR families that refine them: $f_{\text{mono.shape}}$ (qualitative-dynamics shape invariants, refining m_{mono}) and $f_{\text{conv.rate}}$ (method-comparison error-bound partial orders, refining m_{conv}).

3.4.3 PWR corpus placement under canonical symbols.

Provenance and scope of the PWR corpus. The reactor-physics material used here is a PWR MR corpus distilled from standard reactor-physics literature [6, 31] (supplementary S2), not an independent external benchmark. It tests vocabulary coherence: whether reactor-domain MRs can be placed in the same canonical symbol system used throughout the paper. External-transfer evidence (a PARCS V&V or IAEA-TECDOC-class catalogue) and the reactor-side analogue of the commons-math cross-codebase pilot are committed as follow-up (supplementary S4/S5).

The PWR corpus contains five recurring property groups: conservation/invariance, monotonicity, convergence, qualitative trajectory shape, and method-accuracy/bounding. We use these groups only as source material for canonical placement, not as a second MetaPattern vocabulary. Table 2 maps each reactor property group to either one of the five MetaPatterns or a named MR family in Table 1. The same placement also exposes two reactor structures, adjoint reciprocity and collisionless reversibility, that belong to the canonical grid even when a compact engineering MR set does not isolate them.

Table 2. Reactor-physics property groups are placed under canonical NOETHER symbols.

Reactor property group	Canonical symbol	Placement
Conservation / invariance	$G \mapsto f_{\text{inv.con}}$	Conserved quantities are the Noether image of the symmetry MetaPattern m_{inv} .
Monotonicity	$O_{\leq} \mapsto f_{\text{mono.stat}}$	Order-preserving perturbations instantiate the static-order family.
Convergence	$\mathcal{L}^* \mapsto f_{\text{conv.lim}}$	Mesh, time-step, or fixed-point refinement instantiates the convergence family.
Qualitative trajectory shape	$O_{\leq} \mapsto f_{\text{mono.shape}}$	Shape constraints are order relations on trajectories, not a separate MetaPattern.
Method accuracy / bounding	$\mathcal{L}^* \mapsto f_{\text{conv.rate}}$	Method comparisons are convergence-family accuracy-order relations.
Adjoint reciprocity	$T^* \mapsto f_{\text{adj.dual}}$	Bilinear-response identities instantiate the adjoint-duality family.
Collisionless reversibility	$\mathcal{T}_{\text{rev}}^* \mapsto f_{\text{rev.traj}}$	Reversible transport instantiates the trajectory-reversal family.

For example, eight of the eighteen supplementary PWR MRs instantiate G , $O_{\leq}$, or $\mathcal{L}^*$ directly; six instantiate the family-layer refinements $f_{\text{mono.shape}}$ and $f_{\text{conv.rate}}$; and three instantiate $f_{\text{adj.dual}}$ or $f_{\text{rev.traj}}$. The last MR (mixed temperature coefficient under boron concentration) is correctly left outside the current decomposition and becomes the candidate metric-stability/second-order-interaction extension discussed in Appendix D. The gain is canonical placement: reactor-specific property groups are read through the same MetaPattern and MR-family symbols used for equivariant ML, relational optimisers, and the empirical evidence tables.

Element-wise correspondence. Supplementary S2 traces representative reactor-physics MRs from the standard literature [6, 31] to their NOETHER placement, including source equations and

Translate templates. The main text keeps only the property-group mapping because Table 2 is the level at which the symbol system is introduced.

Noether-style derivation and specialisations. Appendix A gives a worked Noether-style derivation of m_{adj} from the Boltzmann bilinear form $\mathcal{F}[\phi, \phi^\dagger]$, derives the adjoint-reciprocity MR ρ_{adj} , and specialises the MetaPattern set to neutron transport, diffusion, and burnup solvers.

3.4.4 Summary. NOETHER applied to $\mathcal{A}_{\text{Boltz}}$ deductively produces the same five-MetaPattern grid used elsewhere in the paper and populates it with reactor-specific MR families. All placements correspond to invariants independently validated in standard reactor-physics literature (symmetry-of-criticality and adjoint-perturbation theory [6], cross-section monotonicity and collisionless time-reversal [31], adjoint-flux treatment [51]). The contribution is therefore not de-novo discovery; these invariants are textbook-canonical. The contribution is the algebraic warrant for treating them as a uniform MetaPattern/MR-family grid. The next section turns to a more demanding case: a domain in which such an MR corpus does not yet exist.

3.5 Cross-domain demonstration: equivariant machine learning

3.5.1 The equivariant-ML program family and its operator algebra. Equivariant neural networks impose by architectural construction that certain symmetries of the input induce predictable transformations of the output [8, 14, 29, 55]. The program family $\mathcal{F}_{\text{equi}}$ comprises classifiers satisfying $f(g \cdot \mathbf{x}) = \rho(g) \cdot f(\mathbf{x})$ for $g \in G = \text{SO}(3) \times \mathfrak{S}_n$. Decomposed along the eight structural components: $G = \{G_{\text{equi}}\}$, $O_{\leq} = \{O_{\leq}^{\text{train}}\}$, $T^* = \{T_{\text{att}}^*\}$, $\mathcal{T}_{\text{rev}}^* = \{\mathcal{T}_{\text{seq}}\}$, $\mathcal{L}^* = \{\mathcal{L}_{\text{train}}, \mathcal{L}_{\text{depth}}, \mathcal{L}_{\text{dim}}\}$, $\mathcal{D}^* = \emptyset$ (for feedforward classifiers), $\mathcal{E}^* = \emptyset$ within a single architecture, $\mathcal{B}_{\text{rel}}^* = \emptyset$ (no idempotent-semiring rewriting on equivariant-classifier outputs).

3.5.2 Running CONSTRUCT-MP on $\mathcal{A}_{\text{equi}}$. CONSTRUCT-MP returns

$$\mathbb{M}(\mathcal{A}_{\text{equi}}) = \{m_{\text{inv}}^{\text{eq}}, m_{\text{mono}}^{\text{eq}}, m_{\text{adj}}^{\text{eq}}, m_{\text{rev}}^{\text{eq}}, m_{\text{conv}}^{\text{eq}}\}.$$

The labels mirror $\mathbb{M}(\mathcal{A}_{\text{Boltz}})$ but the content is domain-specific. This checks the transfer claim: once $\mathcal{A}_{\text{equi}}$ is specified, the same downstream construction produces domain-specific MR families without a reactor-physics corpus. It is not an empirical ML validation.

Cross-corroboration from the equivariant-ML literature. The five derived MetaPatterns correspond to invariants independently validated in the literature: $m_{\text{inv}}^{\text{eq}}$ (symmetry invariance) is the design principle of group-convolutional neural networks (G-CNNs) [14], E(3)-equivariant graph neural networks (EGNNs) [44], and Tensor-Field Networks [55]; $m_{\text{adj}}^{\text{eq}}$ (self-adjoint attention duality) is exploited by the SE(3)-Transformer’s Clebsch–Gordan attention [18]; $m_{\text{rev}}^{\text{eq}}$ relates to reversible networks [19]; $m_{\text{conv}}^{\text{eq}}$ is the implicit invariant of training-curve refinement studies [8]; and gauge-equivariant extensions are catalogued in [13]. NOETHER’s contribution is the uniform algebraic warrant: the same downstream construction derives these MetaPatterns from $\mathcal{A}_{\text{equi}}$ ’s decomposition without re-running the empirical induction behind each result.

3.5.3 End-to-end derivation: a concrete MR for SE(3)-equivariant point-cloud classification. To illustrate the framework’s generative use, we trace a complete derivation from algebra to executable MR.

Step 1. System under test. Let $f : \mathbb{R}^{n \times 3} \rightarrow \Delta^{C-1}$ be a point-cloud classifier mapping n three-dimensional points to a probability distribution over C classes.

Step 2. Distil $\mathcal{A}_{\text{equi}}$. The relevant MetaPatterns are $G = \text{SO}(3) \times \mathfrak{S}_n$ and $\mathcal{L}^*$ for training-size and depth limits.

Step 3. Run CONSTRUCT-MP, select an invariant. Within the G MetaPattern, fix attention on the rotation invariant: $f(R \cdot \mathbf{x}) = f(\mathbf{x})$ for all $\mathbf{x}$ and $R \in \text{SO}(3)$.

Step 4. Translate the invariant into an executable MR.

$$\rho_{\text{rot}} : \quad \forall R \in \text{SO}(3), \quad \|f(R \cdot \mathbf{x}) - f(\mathbf{x})\|_{\infty} \leq \tau, \quad (2)$$

with $\tau = 10^{-4}$ for fp32 architectures.

Listing 1. Executable MR ρ_{rot} for SE(3)-equivariant classifier.

```

Step 5. Generate an executable test.
1 import numpy as np
2 from scipy.spatial.transform import Rotation
3
4 def test_rotation_invariance(model, point_cloud, num_samples=100, tau=1e-4):
5     """Executable MR rho_rot for SE(3)-equivariant classifier."""
6     p_original = model.predict(point_cloud)
7     failures = []
8     for _ in range(num_samples):
9         R = Rotation.random().as_matrix()
10        rotated = point_cloud @ R.T
11        p_rotated = model.predict(rotated)
12        deviation = np.max(np.abs(p_original - p_rotated))
13        if deviation > tau:
14            failures.append((R, deviation))
15    return failures
    
```

Step 5b. A Noether-style reading of ρ_{rot} . The same methodological move that produced m_{adj} in §A.0.1 produces ρ_{rot} here as a concrete derivation. The classifier f is, in equivariant-network architectures by construction, invariant under the action of $G = \text{SO}(3)$ on its input: $f(R \cdot \mathbf{x}) = f(\mathbf{x})$ for all $R \in G$. The Lie algebra $\mathfrak{g} = \mathfrak{so}(3) = \text{span}\{L_x, L_y, L_z\}$ is the infinitesimal generator of this symmetry. The associated Noether-style invariant is the constancy of the classifier output along orbits of G : $\frac{d}{d\theta} f(\exp(\theta L_a) \cdot \mathbf{x}) = 0$ for each generator L_a , $a \in \{x, y, z\}$, which is the Lie-algebraic statement that the directional derivative of f along every orbit-tangent direction vanishes. CONSTRUCT-MP's Step 1 extracts this invariant from the G MetaPattern of $\mathcal{A}_{\text{equi}}$; Translate converts it into the executable MR of equation (2) above. The MR is therefore not an arbitrary catalogue entry: it is the Noether-style consequence of the $\text{SO}(3)$ symmetry that the architecture imposes by construction.

Step 6. Pattern coverage status. The MR ρ_{rot} above, together with ρ_{perm} and ρ_{train} derived in supplementary S9 (Appendix D), populates three of the five non-empty MetaPatterns of $\mathbb{M}(\mathcal{A}_{\text{equi}})$. We refrain from reporting a percentage coverage figure: with so few non-empty MetaPatterns and a denominator subject to the structural sufficiency hypothesis (Hypothesis 1), the raw numerator is more informative than a percentage. We instead derive in the following two subsections two MRs from the remaining non-empty MetaPatterns, T^* (self-adjoint) and $\mathcal{T}_{\text{rev}}^*$ (time-reversal), chosen specifically because they are not standard practice in equivariant-ML testing and therefore probe whether NOETHER's transfer claim extends beyond well-known invariances such as ρ_{rot} .

3.5.4 An adjoint-attention duality MR (ρ_{adj}). The T^* MetaPattern in $\mathcal{A}_{\text{equi}}$ contains the attention-kernel symmetriser T_{att}^* , whose attention-trace invariant $\iota_{\text{att}} = \text{Tr } A(\cdot, \cdot)$ Translate carries to a forward-pass-only continuous-integration (CI)-time MR:

$$\rho_{\text{adj}} : \quad \left| \text{Tr } A(\mathbf{x}^{(1)}, \mathbf{x}^{(2)}) - \text{Tr } A(\mathbf{x}^{(2)}, \mathbf{x}^{(1)}) \right| \leq \tau_{\text{adj}}, \quad (3)$$

for two clouds presented in opposite query/key roles ($\tau_{\text{adj}} = 10^{-4}$ for fp32), executable on any equivariant transformer exposing the attention bilinear form [18, 55]. To our knowledge ρ_{adj} is not catalogued as a standard MR for equivariant attention [2, 47, 60]; NOETHER's contribution is the algebraic warrant for treating it as a T^* MetaPattern member distinct from $m_{\text{inv}}^{\text{eq}}$. A debug-time Gram-matrix variant is in supplementary S1.

3.5.5 A training-trajectory time-reversal MR ($\rho_{\text{train-rev}}$). The $\mathcal{T}_{\text{rev}}^*$ MetaPattern in $\mathcal{A}_{\text{equi}}$ is non-empty for stochastic-gradient-descent (SGD) trajectories under the continuous-time-flow view of stochastic optimisation [9]: vanilla SGD is time-reversible to $O(\eta^2)$, and Translate carries the round-trip identity to a debug-time MR

$$\rho_{\text{train-rev}} : \quad \|\theta_T - \tilde{\theta}_T^{(\text{round-trip})}\|_2 \leq c \eta^2 T, \quad (4)$$

comparing T forward SGD steps against a forward-inverse-forward round trip on the same mini-batch sequence. It *fails by construction* on momentum-based optimisers (Adam, Lion), as predicted by the framework's account of which architecture-optimiser pairs admit the relation, and surfaces gradient-reversal defects such as wrong-sign loss terms [19]; it is invoked once per training-script change on a vanilla-SGD fixture, not in production CI.

Coverage status, revised. The full NOETHER-derived MR set (Set N) consists of five MRs, one per non-empty MetaPattern of $\mathbb{M}(\mathcal{A}_{\text{equi}})$:

$$\text{Set N} = \{ \rho_{\text{rot}} (G), \rho_{\text{mono}} (O_{\leq}), \rho_{\text{train}} (\mathcal{L}^*), \rho_{\text{adj}} (T^*), \rho_{\text{train-rev}} (\mathcal{T}_{\text{rev}}^*) \}.$$

ρ_{rot} is the rotation-invariance MR of Section 3.5.3; ρ_{adj} and $\rho_{\text{train-rev}}$ are the non-trivial MRs of Sections 3.5.4–3.5.5; ρ_{mono} is the point-density-monotonicity MR for the $O_{\leq}$ MetaPattern (top-1 stability under removal of a small fraction of redundant input points), derived from the $O_{\leq}^{\text{train}}$ operator restricted to inference-time input perturbations; ρ_{train} is the inference-idempotency MR for the $\mathcal{L}^*$ MetaPattern. With all five MetaPatterns populated, $\text{coverage}_{\text{NOETHER}}(\text{Set N}) = 1.0$ by construction, this is what Theorem 1 predicts for a $\mathcal{A}_{\text{equi}}$ -derived MR set. The permutation-invariance MR ρ_{perm} (supplementary S9 (Appendix D)) is an auxiliary G -MetaPattern MR retained as a redundant probe and is excluded from Set N to keep the comparison budget $|N| = |L| = |B| = 5$.

The non-trivial nature of ρ_{adj} and $\rho_{\text{train-rev}}$, both absent from the equivariant-ML MR-testing literature we surveyed, substantiates the claim that NOETHER's transfer is generative rather than nominal.

3.6 A third domain: relational query optimisers

Both the Boltzmann (§3.4) and the equivariant-ML (§3.5) instantiations share a common mathematical core: Lie-group symmetry, self-adjoint duality, and time-reversal involution. This is precisely the core that motivated the curation of Hypothesis 1 in the first place. To test transferability beyond this core, this subsection instantiates NOETHER on a domain whose algebraic skeleton is structurally distinct: the relational query optimiser. Relational algebra is built from operators (selection σ , projection π , join $\bowtie$, union $\cup$, set/bag-difference) whose equivalence classes are governed by an idempotent semiring with a partial order under containment, not by a Lie group. There is no obvious self-adjoint operator and no time-reversal involution. The third instantiation therefore tests whether the framework produces useful MetaPatterns outside its training image.

The query-optimiser program family and $\mathcal{A}_{\text{rel}}$. Let $\mathcal{F}_{\text{rel}}$ be the family of programs that take a SQL query q and a database state D and return a relation $\text{eval}(q, D)$; two queries are equivalent if $\forall D. \text{eval}(q, D) = \text{eval}(q', D)$. The operator algebra $\mathcal{A}_{\text{rel}}$ activates G (join / union permutation), $O_{\leq}$ (selection-strengthening + projection-coarsening monotonicity), $\mathcal{E}^*$ (hash- / merge- / nested-loop

plan equivalence under stated cost models [37, 57]), and $\mathcal{B}_{\text{rel}}^*$ (Definition 1: selection-pushdown $\sigma_p(R \bowtie S) = \sigma_p(R) \bowtie S$ when $\text{attr}(p) \subseteq \text{attr}(R)$; idempotent $\sigma_p \circ \sigma_p = \sigma_p$; constant-folding $\sigma_{1=1}(R) = R$, $R \bowtie \emptyset = \emptyset$). Structural components T^* , $\mathcal{T}_{\text{rev}}^*$, $\mathcal{D}^*$, $\mathcal{L}^*$ are empty under $\mathcal{A}_{\text{rel}}$ in their canonical forms (no inner-product self-adjointness; no time-reversal involution; no qualitative dynamics; no ϵ -limit operator on exact-relational semantics). The full per-structural-component enumeration of $\mathcal{A}_{\text{rel}}$ operators is in supplementary S6 query_optimiser/algebra_breakdown.md.

Running CONSTRUCT-MP on $\mathcal{A}_{\text{rel}}$. Steps 1–4 produce three MetaPatterns from the structural-component subset: $m_{\text{inv}}^{\text{rel}}$ (input-permutation invariance under set semantics), $m_{\text{mono}}^{\text{rel}}$ (selection-strengthening monotonicity), and $m_{\text{cmp}}^{\text{rel}}$ (plan-equivalence under cost-model bounds). The relational query optimiser is the *canonical* instantiation in which $\mathcal{B}_{\text{rel}}^*$ is non-empty: it activates the relational extension that neither Boltzmann reactor physics nor equivariant ML exercises.

Four MRs derived for $\mathcal{A}_{\text{rel}}$.

- (1) $\rho_{\text{join-comm}}$ (from G): $\text{eval}(q_1 \bowtie q_2, D) = \text{eval}(q_2 \bowtie q_1, D)$ on bag semantics.
- (2) $\rho_{\text{select-push}}$ (from $\mathcal{B}_{\text{rel}}^*$): $\text{eval}(\sigma_p(R \bowtie S), D) = \text{eval}(\sigma_p(R) \bowtie S, D)$ when $\text{attr}(p) \subseteq \text{attr}(R)$.
- (3) $\rho_{\text{distinct-idem}}$ (from $\mathcal{B}_{\text{rel}}^*$): $\sigma_p \circ \sigma_p = \sigma_p$ as a query-rewrite identity, checkable through plan-tree comparison.
- (4) $\rho_{\text{plan-equiv}}$ (from $\mathcal{E}^*$): two execution plans for the same query produce identical relations within stated NULL-propagation rules [57].

Position and cross-corroboration. NOETHER’s $\mathcal{B}_{\text{rel}}^*$ instantiation is complementary to prior database-testing lines: random SQL generation [5, 49], automated MR generation [45], formal query-equivalence solvers [38, 57, 68], and differential / mutation testing [3, 17, 67]. It provides an algebraically grounded MetaPattern enumeration whose closure under Translate (Theorem 1) none of those lines establishes. The four derived MRs are not de-novo discoveries: they are canonical query-equivalence identities already validated by formal solvers ($\rho_{\text{join-comm}}$ and $\rho_{\text{plan-equiv}}$ are central to SPES/QED’s Calcite verification [57, 68], $\rho_{\text{select-push}}$ is the standard selection-pushdown rewrite, and $\rho_{\text{distinct-idem}}$ is certifiable by Mohamed et al.’s SMT theory [38]). NOETHER’s contribution is the uniform algebraic warrant: the same downstream construction produces MetaPatterns that exercise the relational extension the other two domains do not activate. A pre-registered protocol comparison on Segura et al.’s IMDb subset is in supplementary S6.

What this third domain establishes, and the Theorem 1’ verdict on $\mathcal{A}_{\text{equi}}$ and $\mathcal{A}_{\text{rel}}$. The instantiation establishes that NOETHER applies outside the Lie-group / self-adjoint / time-reversal core: relational optimisers exercise $\mathcal{B}_{\text{rel}}^*$ where the first two domains do not. The counterexample-search protocol applied to both $\mathcal{A}_{\text{equi}}$ and $\mathcal{A}_{\text{rel}}$ (mirroring §3.7) gives the following verdict. On $\mathcal{A}_{\text{equi}}$: Theorem 1’ is falsified by two pairwise-independent candidates: ρ_{compose} (joint $\text{SO}(3) \times \mathfrak{S}_n$ action on point sets [44], requiring a product-group π -template) and ρ_{gauge} (gauge-equivariance on manifolds [13], requiring a bundle-section π -template parametrised by $\mathfrak{g} \in \mathcal{C}(M, H)$). The two extensions are independent (a product-group π does not absorb a gauge-bundle section); the $\text{SO}(3)$ Lie-algebra structure alone does not force closure on $\mathcal{A}_{\text{equi}}$. On $\mathcal{A}_{\text{rel}}$: a survey of ≥ 10 unverified cases from [57]’s residue (Aggregate×Project, constant-key, Decorrelate, NULL three-valued logic) produces five Theorem 1’ candidates; the primary witness $\rho_{\text{agg-proj}}$ (Calcite’s AggregateExtractProjectRule) requires $\mathcal{B}_{\text{rel}}^*$ plus an aggregation-as-algebra additional MetaPattern. Combined with the five $\mathcal{A}_{\text{PWR}}$ obstructions of Table 3, the three algebra-survey artefacts (theory/equi_thm1prime_search.md, theory/rel_thm1prime_search.md,

theory/translate_extensions.md) identify ten Translate-extension dimensions. Pairwise independence is proved by per-MetaPattern exhaustion on the five PWR obstructions (Appendix D); the five candidate $\mathcal{A}_{\text{equi}} + \mathcal{A}_{\text{rel}}$ dimensions (two specialising PWR-side dimensions to type-distinct primitives, three net relational-side) are asserted independent by inspection, with formal per-dimension exhaustion as follow-up. Remark 3 catalogues six further out-of-decomposition program-family classes.

3.7 A negative instantiation: irreducibly compositional MRs in PWR core simulators

Sections 3.4, 3.5, and 3.6 instantiated NOETHER on three program families (Boltzmann reactor physics, equivariant ML, and relational query optimisers) for which the framework's downstream construction was non-vacuous and produced executable MRs. This subsection instantiates NOETHER on a fourth program family, the PWR core diffusion solver family, with an inverted purpose: rather than demonstrating coverage, we exhibit two specific MRs from the standard PWR safety-analysis literature that the framework's Translate operator cannot reach under any single-component derivation. The two MRs together identify five pairwise-independent structural obstructions in Translate's present signature, jointly recasting Theorem 1' (Conjecture D, Appendix D) from an open conjecture to a falsified statement on a structurally significant operator algebra.

The MRs chosen are not pathological cases. They are core safety-analysis MRs that PWR core simulators are required by regulatory practice and engineering convention to reproduce: non-additivity of control-bank reactivity worth (the algebraic root of rod-bank shadowing and anti-shadowing phenomena) and second-order mixed dependence of k_{eff} on moderator temperature and boron concentration (the standard MTC-vs-boron design curve). The negative instantiation thus uses NOETHER's principal application domain (reactor physics) to test the framework's most ambitious stated claim (algebraic closure over arbitrary single-component-derivable MRs).

Why PWR. We use the PWR core diffusion algebra rather than $\mathcal{A}_{\text{equi}}$ or $\mathcal{A}_{\text{rel}}$ for two reasons: regulatory essentiality (10 CFR 50 and NRC Reg. Guide 1.77 require core simulators to reproduce the two MRs below, so the counterexamples are not contrived) and engineering documentability (the algebra has a published canonical form [6, 31] against which the structural obstructions can be precisely located, which $\mathcal{A}_{\text{equi}}$ and $\mathcal{A}_{\text{rel}}$ do not yet support). Whether they admit analogous Theorem 1' counterexamples is open follow-up.

The PWR core diffusion algebra. Let $\mathcal{F}_{\text{PWR}}$ be the program family of PWR core diffusion solvers (canonical examples: PARCS, SIMULATE-3/5, ANC, SMART). Its operator algebra $\mathcal{A}_{\text{PWR}}$ contains, in addition to the operators of $\mathcal{A}_{\text{Boltz}}$ (Section 3.4.3), the following PWR-specific generators:

- $\mathcal{O}_{\text{rod}}$: discrete control-rod insertion operators, parametrised by rod-bank label g and insertion depth d , generating an additive-on-geometry semigroup under composition. In the steady-state setting we adopt throughout, $\mathcal{O}_{\text{rod}}^A \cdot \mathcal{O}_{\text{rod}}^B$ and $\mathcal{O}_{\text{rod}}^B \cdot \mathcal{O}_{\text{rod}}^A$ act identically on the input space (both produce the same total inserted geometry), so $\mathcal{O}_{\text{rod}}$ is commutative on geometry. The *reactivity-worth functional* on $\mathcal{O}_{\text{rod}}$, however, is not a semigroup homomorphism: $d\rho(A \cup B) \neq d\rho(A) + d\rho(B)$ in general. This non-additivity, not non-commutativity, is what the present subsection exploits.
 - $\mathcal{M}_{C_B}$: continuous boration operators acting on the moderator material composition through the soluble-boron concentration C_B (typical PWR operating range: 0–2000 ppm).
 - $\mathcal{M}_{T_{\text{mod}}}$: continuous moderator-temperature operators acting on the cross-section library through the parametric dependence $\Sigma(T_{\text{mod}})$ (typical PWR operating range: 290–320°C).
- Decomposed along the structural decomposition of Section 3.1.1:

- $G \supseteq \{O_{\text{rod}}\}$ (treated tentatively as a commutative semigroup; the assignment will be shown below to fail);
- $O_{\leq} \supseteq \{\mathcal{M}_{C_B}, \mathcal{M}_{T_{\text{mod}}}\}$ (parameter-monotonicity operators);
- $T^* \supseteq \{-\nabla \cdot D\nabla + \Sigma_a\}$ (the self-adjoint diffusion operator under isotropic scattering; Bell & Glasstone [6] §6.1, Lewis & Miller [31] §4.2);
- $\mathcal{T}_{\text{rev}}^* = \emptyset$ (PWR diffusion is dissipative);
- $\mathcal{L}^*, \mathcal{D}^*, \mathcal{E}^*$ as in $\mathcal{A}_{\text{Boltz}}$ with appropriate restrictions to the diffusion regime;
- $\mathcal{B}_{\text{rel}}^* = \emptyset$ (no idempotent-semiring structure on PWR core states).

We will show that despite this rich structural decomposition, two specific PWR-safety MRs cannot be derived through Translate from any single structural component.

Main proposition: non-additivity of rod-bank reactivity worth.

DEFINITION 11 (DIFFERENTIAL ROD-BANK REACTIVITY WORTH, EXACT FORM). *For a base input $x_0 \in X$ and a rod-bank operator $A \in O_{\text{rod}}$, the (positive-convention) reactivity worth of A is*

$$d\rho(A; x_0) := \frac{1}{k_{\text{eff}}(P(x_0))} - \frac{1}{k_{\text{eff}}(P(O_{\text{rod}}^A \cdot x_0))} > 0,$$

where $k_{\text{eff}}(P(x))$ denotes the dominant eigenvalue of the diffusion operator at configuration x . We write $d\rho(A \cup B; x_0)$ when both banks A and B are inserted simultaneously. Equivalently, in conventional reactor-physics notation, $d\rho(A; x_0) = \rho(x_0) - \rho(O_{\text{rod}}^A \cdot x_0)$ where $\rho = 1 - 1/k_{\text{eff}}$ is the static reactivity. The exact form avoids first-order perturbation-theoretic approximation; the standard adjoint-perturbation reading appears below.

DEFINITION 12 (NON-ADDITIVITY OF ROD-BANK REACTIVITY WORTH, ρ_{nonadd}). *For two control-rod banks $A, B \in O_{\text{rod}}$ with disjoint geometric supports and a base input x_0 , define the mixed-difference functional*

$$\Delta_{AB}(x_0) := d\rho(A \cup B; x_0) - d\rho(A; x_0) - d\rho(B; x_0).$$

The non-additivity metamorphic relation asserts: there exist disjoint-support banks A, B and base input $x_0 \in X$ in the standard PWR operating envelope (typical multi-bank insertion patterns of D, C, B, A control banks at partial insertions) such that

$$\rho_{\text{nonadd}} : |\Delta_{AB}(x_0)| > \tau_{\text{nonadd}},$$

with $\tau_{\text{nonadd}} = 5$ pcm. The tolerance is calibrated to PWR engineering practice: empirical $|\Delta_{AB}|$ for adjacent rod banks ranges over 10^1 – 10^2 pcm [51]; the tolerance $\tau_{\text{nonadd}} = 5$ pcm lies safely above PWR core-simulator iterative convergence tolerances (typically 0.1–1 pcm for k_{eff}) and below the physical signal magnitude. Selection of test bank configurations (A, B) is the user's responsibility; the framework's failure to derive ρ_{nonadd} is independent of the specific tolerance choice.

PROPOSITION 1 (NON-ADDITIVITY IS NOT Translate-REACHABLE ON $\mathcal{A}_{\text{PWR}}$). *Let $\mathcal{A}_{\text{PWR}}$ be the PWR core diffusion algebra above, with structural decomposition $\mathcal{D}(\mathcal{A}_{\text{PWR}})$. For every structural component $s \in \mathcal{D}(\mathcal{A}_{\text{PWR}})$ and every invariant $\iota \in \mathcal{I}_s$, $\text{Translate}(\iota, s) \neq \rho_{\text{nonadd}}$. Equivalently, $\rho_{\text{nonadd}} \notin \text{MR}(\mathcal{A}_{\text{PWR}})$ in the sense of Definition 5.*

The proof, by exhausting the eight structural components against the per-structural-component Translate templates of Table 16, is given in Appendix D.

Supporting proposition: second-order mixed dependence of k_{eff} on T_{mod} and C_B .

DEFINITION 13 (MTC-VS-BORON MIXED-DERIVATIVE MR, $\rho_{\text{MTC-bor}}$). *Let $k_{\text{eff}}(T_{\text{mod}}, C_B; \xi_0)$ denote the dominant eigenvalue of the PWR core diffusion operator as a function of moderator*

temperature T_{mod} and soluble-boron concentration C_B , holding fixed the auxiliary state $\xi_0 = (\text{BU}_0, T_{\text{fuel},0}, \text{geometry, loading pattern, rod-bank position})$. Define the static reactivity

$$\rho_{\text{static}}(T_{\text{mod}}, C_B; \xi_0) := 1 - \frac{1}{k_{\text{eff}}(T_{\text{mod}}, C_B; \xi_0)},$$

expressed in pcm units ($1 \text{ pcm} = 10^{-5}$). The moderator temperature coefficient (MTC) at $(T_{\text{mod}}, C_B; \xi_0)$ is the partial derivative of static reactivity with respect to moderator temperature, in standard PWR engineering convention [6, §10.3], [50, §3.4]:

$$\alpha_{\text{MTC}}(T_{\text{mod}}, C_B; \xi_0) := \left. \frac{\partial \rho_{\text{static}}}{\partial T_{\text{mod}}} \right|_{C_B, \xi_0 \text{ fixed}}, \quad (\text{units: pcm/}^\circ\text{F or pcm/}^\circ\text{C}).$$

The second-order mixed dependence MR asserts: for $(T_{\text{mod}}, C_B; \xi_0)$ in the standard PWR operating envelope at hot-full-power (HFP), all-rods-out (ARO) reference conditions (typical: $T_{\text{mod}} \in [290, 320]^\circ\text{C}$, $C_B \in [0, 2000] \text{ ppm}$, $\text{BU}_0 \in [0, 50] \text{ GWd/tU}$, full-power $T_{\text{fuel},0}$), the mixed second partial derivative satisfies

$$\rho_{\text{MTC-bor}} : \left| \frac{\partial^2 \rho_{\text{static}}}{\partial T_{\text{mod}} \partial C_B} \right| > \tau_{\text{MTC-bor}},$$

with $\tau_{\text{MTC-bor}} = 0.01 \text{ pcm/}^\circ\text{F/ppm}$ (equivalently $\sim 1.8 \times 10^{-2} \text{ pcm/}^\circ\text{C/ppm}$). The tolerance is calibrated to PWR engineering practice: the empirical value of $\partial \alpha_{\text{MTC}} / \partial C_B$ in Westinghouse/Framatome PWR designs ranges over $0.02\text{--}0.04 \text{ pcm/}^\circ\text{F/ppm}$ at BOC-to-EOC cycle conditions [30, 50] (Stacey, §3.4; Lamarsh–Baratta, §8.3); the tolerance $\tau_{\text{MTC-bor}} = 0.01 \text{ pcm/}^\circ\text{F/ppm}$ lies safely below this physical magnitude and well above PWR core-simulator differential-perturbation noise (typically $\sim 10^{-3} \text{ pcm/}^\circ\text{F/ppm}$ for converged eigenvalue calculations).

Equivalent formulation in terms of MTC.. Definition 13 is equivalent to asserting

$$\left| \frac{\partial \alpha_{\text{MTC}}(T_{\text{mod}}, C_B; \xi_0)}{\partial C_B} \right| > \tau_{\text{MTC-bor}}$$

at HFP, ARO conditions. Physically, this captures the well-established PWR design property that MTC becomes monotonically more negative as C_B decreases from BOC values ($\sim 1500 \text{ ppm}$) to EOC values ($\sim 0 \text{ ppm}$). At HFP operating conditions, MTC is regulated to be $\leq 0 \text{ pcm/}^\circ\text{F}$ across the entire operating range per 10 CFR 50 Appendix A General Design Criterion 11 [56]; the slope $\partial \alpha_{\text{MTC}} / \partial C_B$ governs how rapidly MTC moves toward more negative values as boron is depleted over the cycle, with typical magnitudes ranging from near-zero (at BOC, high boron) to -30 to $-50 \text{ pcm/}^\circ\text{F}$ (at EOC, near-zero boron). At hot-zero-power (HZIP) conditions, BOC MTC may approach zero or be slightly positive within the analytical envelope (a regime relevant to startup physics testing but not to power-operation safety analysis); HZIP MTC is bounded by separate Technical Specifications limits.

The strength of $\partial \alpha_{\text{MTC}} / \partial C_B$ is the engineering target of *MTC-vs-boron concentration curve* calculations performed for every PWR cycle reload, and is governed by the competition between three physical mechanisms [50, §3.4]:

- Reduced moderation:** $T_{\text{mod}} \uparrow \Rightarrow \text{moderator density} \downarrow \Rightarrow \text{neutron moderation reduced} \Rightarrow \text{thermal-flux fraction} \downarrow \Rightarrow \text{fission rate} \downarrow$. Contributes a *negative* term to MTC.
- Boron poison evacuation:** at high C_B , $T_{\text{mod}} \uparrow \Rightarrow \text{moderator density} \downarrow \Rightarrow \text{boron number density} \downarrow$ (since boron is dissolved in the moderator) $\Rightarrow \text{boron absorption} \downarrow$. Contributes a *positive* term to MTC, partially cancelling (a). This term is proportional to C_B and vanishes as $C_B \rightarrow 0$.
- Spectrum hardening and ^{238}U resonance enhancement:** $T_{\text{mod}} \uparrow \Rightarrow \text{reduced moderation} \Rightarrow \text{harder neutron spectrum} \Rightarrow \text{enhanced } ^{238}\text{U} \text{ resonance absorption (Doppler-weighted by the fuel temperature)} \Rightarrow \text{neutron loss} \uparrow$. Contributes a further *negative* term to MTC. This term

is small for low-enriched UO_2 but becomes significant for MOX fuels and high-enrichment ($> 5\%$) UO_2 .

At high C_B the partial cancellation from (b) is strong; mechanisms (a) and (c) are partially offset and MTC magnitude is small. At low C_B , mechanism (b) vanishes; mechanisms (a) and (c) dominate and MTC becomes strongly negative. The mixed second derivative $\partial^2 \rho_{\text{static}} / (\partial T_{\text{mod}} \partial C_B)$ measures the rate at which the boron-mediated cancellation is removed as C_B decreases.

PROPOSITION 2 (MTC-VS-BORON MIXED DEPENDENCE IS NOT Translate-REACHABLE). *Let $\mathcal{A}_{\text{PWR}}$ be the PWR core diffusion algebra above. For every structural component $s \in \mathcal{D}(\mathcal{A}_{\text{PWR}})$ and every invariant $\iota \in \mathcal{I}_s$, $\text{Translate}(\iota, s) \neq \rho_{\text{MTC-bor}}$. Equivalently, $\rho_{\text{MTC-bor}} \notin \text{MR}(\mathcal{A}_{\text{PWR}})$ in the sense of Definition 5.*

The proof, by reducing the obstruction to the high-order-mixed-difference structure of $\rho_{\text{MTC-bor}}$ and verifying that no per-structural-component π template captures such structure, is given in Appendix D.

Five independent structural obstructions. The two propositions are independent in the strong sense that no single extension of NOETHER's Translate repairs both:

Table 3. Five independent obstructions delimit Translate's current signature.

Proposition	Failure mode	Required extension to Translate
1 (non-additivity)	Output is an algebraic-spectrum quantity (k_{eff} as an eigenvalue, not in $\mathcal{Y}$)	Operator-spectrum output relations on π 's codomain
1 (non-additivity)	Worth functional is non-additive (failure of semigroup homomorphism)	Homomorphism-failure π -template alongside equivariance / monotonicity / self-adjointness
1 (non-additivity)	Adjoint weighting function $\phi_X^\dagger$ varies with operator history X	Configuration-indexed adjoint structure on T^*
2 (mixed derivative)	MR is a non-zero second-order mixed partial derivative, not a first-order relation	Higher-order mixed-difference π -templates (currently all π templates are first-order)
2 (mixed derivative)	MR involves two independent parameter directions (T_{mod}, C_B) jointly, not chained or single-direction	Two-direction joint parametric dependence beyond the single- θ partial order of Definition 3

The five rows in this table identify *five pairwise-independent structural features* of Translate that are absent in Definition 4. Each row, considered alone, would require a specific extension of Definition 4; no single extension covers any two rows simultaneously. The structural decomposition (Hypothesis 1) is therefore not the only constraint on the framework's reach; the *shape of Translate itself* is.

The two MRs ρ_{nonadd} and $\rho_{\text{MTC-bor}}$ are also *physically independent*: they probe disjoint physical mechanisms (rod-induced adjoint distortion vs. moderator-poison density coupling), and a PWR core simulator could pass one MR while failing the other (and vice versa). They are therefore complementary verification targets, not duplicates of a single underlying property.

What this subsection establishes and does not establish. Established. Theorem 1' (Conjecture D) is false on $\mathcal{A}_{\text{PWR}}$: there exist two specific MRs, each empirically realised on every conforming PWR core simulator and each documented in standard PWR safety-analysis literature and regulatory

guidance, that are formulable over $\mathcal{A}_{\text{PWR}}$'s operators but not in $\text{MR}(\mathcal{A}_{\text{PWR}})$ in the sense of Definition 5. The two MRs identify five pairwise-independent structural obstructions in Translate's present signature.

Not established. That Theorem 1 itself fails. Theorem 1's closure result is over $\text{MR}(\mathcal{A}_{\text{PWR}})$ as defined by Definition 5; the two MRs of this subsection lie *outside* that set, so they are out-of-scope for Theorem 1 and consistent with it. The proper characterisation is that Theorem 1 is a substantially weaker statement than Theorem 1' pretended to be, and the gap is now exhibited concretely with two independent witness MRs.

Not established. That a Composite-Translate extension of NOETHER would absorb these two MRs while preserving Theorem 1's closure and Theorem 2's polynomial-time constructibility. This is the principal open problem the negative instantiation leaves to follow-up work. The five obstructions of Table 3 are pairwise independent, so any candidate extension must address them jointly rather than sequentially.

4 Experiments

This section defines the evidence protocol used to evaluate NOETHER as an operator-algebraic MR-identification method. The unit of analysis is a canonical MR family $f_{\text{parent.child}}$ and, where the distinction matters, its parent MetaPattern. The experiments are not designed to estimate average fault-detection superiority; they measure coverage of the canonical design space, explanatory warrant, cross-domain recurrence, and executability/blind-spot predictions.

RQ1: coverage of the canonical design space. Can NOETHER identify MetaPatterns or MR families that are absent from expert MR sets? We answer this with binary canonical coverage: an expert corpus covers a MetaPattern or MR family only when it contains at least one explicit, applicable MR in that row; NOETHER covers it only when CONSTRUCT-MP derives the row from the program-family operator algebra and the program interface exposes the needed controls and observations.

RQ2: origin and boundary explanation. How does NOETHER differ from search-based and category-based MR generation? We compare the warrant attached to each candidate: whether the method explains the algebraic source of the relation, states where the relation stops applying, and collapses redundant variants into a maintainable canonical family.

RQ3: cross-domain and real-fault recurrence. Do the same canonical MR families recur across structurally different domains and in real defects? We trace MR families across Boltzmann/reactor physics, equivariant ML, relational query optimisers, held-out numerical libraries, and mined in-the-wild scientific-computing faults.

RQ4: executability and predicted blind spots. Can selected algebra-derived MR families be executed, and do their successes and failures match the Invariance-Blindness prediction? Mutation, DeepCrime-style, real-fault, and head-to-head results are retained for this purpose. They are secondary evidence for executability, complementarity, and predicted blind spots; they do not decide an average-superiority claim.

Subjects, metrics, methods, and baselines. We use Set N for NOETHER-derived MRs, Set L for the LLM-prompted baseline, Set B for the literature baseline, Set G for GenMorph, and Set M for the METRIC+ held-out baseline. For RQ4, D1 denotes algebra-disrupting mutants and D2 denotes algebra-preserving mutants; PIT denotes the PIT mutation-testing tool [15]. Table 4 states the experimental design in the conventional vocabulary of research questions, subjects, measurements, protocol, and comparators.

Table 4. The evaluation protocol states the subjects, metrics, methods, and comparators for RQ1–RQ4.

RQ	Subjects	Metrics	Method	Comparator
RQ1	SACOS, SPARK, LOCUST, PWR corpus, held-out numerical libraries	Binary MetaPattern/MR-family coverage	Map each explicit MR to its canonical MetaPattern or MR family	Expert MR sets
RQ2	MR-Scout, GenMorph, LLM-prompted MR generation, METRIC/METRIC+	Origin, boundary, redundancy, maintainability	Compare explanatory warrants rather than pooled fault-detection rates	Search/category enumeration
RQ3	Boltzmann/reactor, equivariant ML, relational optimisers, real scientific-computing faults	Cross-domain and real-fault recurrence of canonical MR families	Trace each MR family to the algebraic component that makes it valid	Domain-specific derivations and in-the-wild fix commits
RQ4	EGNN mutations, DeepCrime-style mutations, PIT/GenMorph substrate, IBT harness	Executability, unique detections, D1/D2 strata, oracle complementarity	Run selected derived MRs and compare predicted blind spots/complementarity	Set L, Set B, Set G, and a neutral differential oracle

Baseline selection. The evaluation uses one comparator for each relevant state-of-the-art (SOTA) mechanism. GenMorph’s genetic-programming (GP)-evolved Set G [2], run at its published 30-min budget, represents the genetic-programming arm. A multi-vendor $\times$ multi-temperature LLM ensemble (Set L; §5.2, supplementary S4) represents prompt-based MR generation. MR-Scout [60] represents the mining arm and is contextualised rather than re-executed because its required input, an existing test corpus, is structurally absent on the stand-alone mathematical SUTs of §5.3.3. METRIC+ [53] is treated as a category-enumeration scaffold rather than as an executable pipeline; AutoMT [34] and GPTMR [65] target traffic-rule domains outside the operator-algebraic substrate. Set B is a literature-MR baseline [34, 47, 48] restricted to point-cloud-classifier-applicable MRs.

5 Results and Discussion

This section reports the evidence in the order implied by the research questions and then discusses the implications for MR identification, expert knowledge, search-based generation, and program-family testability. Mutation and head-to-head material is interpreted within the RQ4 executability and blind-spot scope declared in Section 4.

Reading guide for the results. RQ1 is carried by the canonical coverage material: expert MR sets are compared with NOETHER by binary MetaPattern/MR-family coverage, not by the number of MRs or the number of faults exposed. RQ2 is carried by the GenMorph/METRIC+/LLM material as a complementarity analysis: search and category-enumeration methods can produce useful candidates, while NOETHER supplies algebraic origin and boundary. RQ3 is carried by the cross-domain and real-fault traces: the same MR families recur across Boltzmann/reactor physics, equivariant ML, relational optimisers, and held-out numerical or industrial program families. RQ4 is carried by the mutation, DeepCrime-style, head-to-head, and IBT checks; these document executability, category-specific behavior, complementarity, and predicted blind spots.

5.1 Primary MR-identification evidence

Table 5 is the load-bearing evidence for RQ1–RQ3. It uses the binary counting rule defined in Section 4: a structural component is covered only when the source contains at least one explicit, applicable MR class from that structural component. A mathematical structure that exists in a

governing equation but cannot yet be made executable through the program interface is recorded as a limitation, not as coverage.

Table 5. Primary evidence matrix combines the RQ1–RQ3 coverage, explanation, and recurrence evidence.

RQ	Evidence	Main finding	Scope of claim
RQ1	SACOS/SPARK/LOCUST and held-out numerical-library corpora under binary canonical coverage	Expert industrial relations explicitly cover $O_{\leq}$; when interfaces expose the relevant operations, NOETHER covers G , T^* , $\mathcal{L}^*$, $\mathcal{E}^*$, and $G : f_{\text{inv.con}}$ in addition to $O_{\leq}$.	Broader canonical design-space coverage, not higher mutation score.
RQ2	MR-Scout, GenMorph, LLM-prompted generation, METRIC/METRIC+ comparison	Search/category enumeration supplies useful executable candidates but usually leaves origin, boundary, and redundancy as empirical artefacts; NOETHER ties each retained class to a structural component and a source identity or inequality in $\mathcal{A}_P$.	Algebraic warrant and boundary explanation, not measured maintenance-cost reduction.
RQ3	Boltzmann/reactor, equivariant ML, relational optimisers, and real scientific-computing faults	Shared families recur across domains: $G : f_{\text{inv.eqv}}$, $G : f_{\text{inv.con}}$, $T^* : f_{\text{adj.dual}}$, $O_{\leq} : f_{\text{mono.stat}}$, $\mathcal{L}^* : f_{\text{conv.lim}}$, and $\mathcal{L}^* : f_{\text{conv.repr}}$.	Cross-domain recurrence of the canonical symbol system, not proof that the list is absolutely complete.

Taken at the intended level, Table 5 shows that NOETHER identifies MR classes by making algebraic origin and boundary explicit, and that it covers a broader structural-component design space than the expert industrial corpora available to us. It does not establish higher mutation score, average fault-detection superiority, or lower maintenance cost in practice. Its practical case is auditability, transferability of the symbol system, and predictable blind spots once the upstream algebra has been curated.

5.2 Secondary executability check: equivariant-ML case study

The derivations in Sections 3.5.3–3.5.5 establish that NOETHER produces concrete, executable MRs in the equivariant-ML setting. This subsection is retained only as a secondary executability and sanity-check study. It shows that selected derived MR classes can be run against a real model and can expose the defect category they were constructed to probe; it is not evidence for the paper’s main MR-identification claim and it is not used to claim average-case fault-detection superiority. The study is a case study in the strict sense: a single model, a small mutation set, and three MR sets.

Setup. We test five NOETHER-derived MRs (Set N, one per non-empty MetaPattern of $\mathcal{A}_{\text{equi}}$: ρ_{prot} , ρ_{mono} , ρ_{train} , ρ_{adj} , $\rho_{\text{train-rev}}$) against an LLM-prompted baseline (Set L, GPT-4, no structural cue) and a literature baseline (Set B), on a compact E(3)-equivariant graph neural network [44] (EGNN; two layers, 5,189 parameters, validation accuracy 0.93) under $N_{\text{mut}} = 20$ defects in four categories [55]: (i) wrong-sign loss, (ii) equivariance break, (iii) precision degradation, (iv) gradient-reversal sign (5 each). For the RQ4 case-study check, we pre-register two expectations: Set N has full canonical coverage by construction whereas Sets L and B do not, and Set N uniquely detects the category-(iv) defect via $\rho_{\text{train-rev}}$. Architecture, scripts, and the full 20-mutation outcome matrix are in supplementary S3.

Results. Set N detects 7/20 mutations, Set L detects 2/20, and Set B detects 0/20; Set N’s structural coverage is 1.00 versus 0.20 for Sets L and B, and its five unique detections are exactly the category-(iv) gradient-reversal-sign mutations (5/5), while all three sets miss the category-(i) wrong-sign-loss mutations (0/5). The full row-level outcome matrix (one row per (mr, mutation) pair, 300 rows

in total) is provided as supplementary material S3, together with the runner script (`runner.py`) and the analysis script (`analysis.py`) that produce these numbers deterministically from the trained checkpoint. The model under test is an E(3)-equivariant graph neural network [44] (5-class procedural point-cloud classifier; 5,189 parameters; trained on Apple M1 Pro / Metal Performance Shaders (MPS); validation accuracy 0.93 after 25 epochs).

Verdicts. The coverage expectation holds by construction ($\text{coverage}_{\text{NOETHER}} 1.00$ for Set N vs 0.20 for Sets L and B), read as a structural-prior diagnostic, not a fault-detection metric. The unique-detection expectation is consistent and construct-validity-controlled: Set N uniquely detects all five category-(iv) mutations, every time via $\rho_{\text{train-rev}}$, which Sets L and B miss for lack of an algebraic warrant on the $\mathcal{T}_{\text{rev}}^*$ MetaPattern; the asymmetry is significant for N-vs-B after Holm–Bonferroni (McNemar $p = 0.047$) but not for N-vs-L. Because the mutation set covers one category per non-empty MetaPattern, this exhibits construct validity of $\rho_{\text{train-rev}}$ rather than averaged superiority (full pairwise statistics in supplementary S3).

A framework boundary, not an edge case. The most informative cell is that *no* MR set detects category-(i) wrong-sign mutations: a sign-flipped head still satisfies rotation and permutation invariance, inference idempotency, the adjoint identity, and density monotonicity. The structural decomposition contains no *label-consistency* component and is therefore blind to this class by construction. This is the label-consistency out-of-scope class of Remark 3, a documented boundary rather than a defect; the tolerance sweep for the category-(iii) precision cell is in supplementary S3.

5.2.1 A small DeepCrime-style real-fault pilot. A small DeepCrime-style pilot ($n = 5$ real-fault-style mutation operators on the trained checkpoint [24]) detects 2/5 for Set N versus 0/5 for Sets L and B (Wilson 95% CI [0.12, 0.77]; McNemar exact $p = 0.500$, underpowered for $\alpha = 0.05$). The two detections (ρ_{train} firing on head-magnitude collapse) are descriptive evidence consistent with the $\mathcal{L}^*$ -MetaPattern prediction; the three undetected mutations (activation change, bias removal, weight re-init) motivate the parameter-distribution candidate-additional MetaPattern of Remark 3. The pilot’s role is to motivate, not confirm; the comparative-evaluation protocol, the detection mechanism, and case-study threats are in supplementary S3/S4.

Real-bug evaluation (realised): applicability corroboration across four domains. To address threat (c) with a fault distribution the framework was *not* designed against, we mined in-the-wild faults from four third-party scientific-computing libraries, one per paper SUT domain: SciPy (`pde_numerical`), PySCF (`quantum_chemistry`), OpenMC (`reactor_physics`), and DeepXDE (`pde_sciml`, a third-party physics-informed-neural-network library). Each fault is a maintainer fix commit identified by a real SHA; for each we reproduced the pre-fix (FIRED) and post-fix (HELD) behaviour by released-to-released pip/conda installation, or, where the fix was unreleased, by source compilation of the parent and fix commits, then ran the corresponding MR-family check on each. This study evaluates *applicability*, that is, whether the algebra-derived MR families correspond to faults that actually occurred, not fault-revealing *effectiveness*: each populated cell is an instance of the Invariance-Blindness reading (Theorem 3) on a real defect, and no superiority claim is made here. Under RQ3, this realised corpus supplies real-fault recurrence evidence: the same canonical MR families recur across four independent libraries, witnessing cross-domain sharing on deployed defects rather than on derivation traces alone.

Coverage. Twenty-one in-scope faults populate 8 of the 10 MR families (Table 1) across the four SUT domains: $f_{\text{inv.eqv}}$ in all four (OpenMC `Surface.normalize` and `RotationalPeriodicBC`, PySCF D_{2h} orbital-symmetry labelling, DeepXDE `periodic-point`, and the numerically marginal SciPy `fht Hermitian`); $f_{\text{inv.con}}$ in PySCF (`electron count under smearing`) and DeepXDE (`Neumann`

mass flux); $f_{\text{adj.self}}$ in SciPy (eigh driver-invariance and complex-symmetric solve/inv) and DeepXDE (forward-mode Hessian symmetry, reachable only off the default path); $f_{\text{adj.dual}}$ in OpenMC (iterated-fission-probability adjoint-weighted kinetics); $f_{\text{mono.stat}}$ in SciPy (Akima two-point linearity), OpenMC (Chebyshev rational approximation method (CRAM) depletion non-negativity), and DeepXDE (float32 boundary detection); $f_{\text{conv.lim}}$ in all four (SciPy LSODA dense output, PySCF symmetry-adapted direct inversion in the iterative subspace (DIIS), OpenMC tally-trigger binding, DeepXDE fixed-collocation training); $f_{\text{conv.rate}}$ in SciPy (composite-Simpson even-point observed order $4 \rightarrow 3$); and $f_{\text{conv.repr}}$ in SciPy (banded-versus-full Jacobian) and OpenMC (Message Passing Interface (MPI) tally reduction). Table 6 summarises the family-by-domain matrix; the per-fault SHAs, versions, reproduction scripts, and FIRED/HELD numbers are in supplementary S5 (real_bugs/).

Table 6. Real faults populate eight of ten MR families across four scientific-computing libraries.

MetaPattern	MR family (mode)	SciPy pde_num.	PySCF quantum_chem.	OpenMC reactor_phys.	DeepXDE pde_sciml
G	$f_{\text{inv.eqv}}$ equivariance (I)	Δ	✓	✓✓	✓
G	$f_{\text{inv.con}}$ conservation (I)	n/a	✓	n/a	✓
T^*	$f_{\text{adj.self}}$ self-adjoint (I/M)	✓✓	$\times^\dagger$	n/a	Δ
T^*	$f_{\text{adj.dual}}$ adjoint duality (M)	n/a	n/a	✓	n/a
$\mathcal{T}_{\text{rev}}^*$	$f_{\text{rev.traj}}$ trajectory reversal (I)	$\times$	$\times$	$\times$	$\times$
$O_{\leq}$	$f_{\text{mono.stat}}$ static order (I)	✓	$\times^\dagger$	✓	✓
$O_{\leq}$	$f_{\text{mono.shape}}$ dynamic shape (I)	$\times$	$\times$	$\times$	$\times$
$\mathcal{L}^*$	$f_{\text{conv.lim}}$ convergence (I)	✓	✓	✓	✓
$\mathcal{L}^*$	$f_{\text{conv.rate}}$ accuracy-order (M)	✓	n/a	n/a	n/a
$\mathcal{L}^*$	$f_{\text{conv.repr}}$ representation-inv. (M)	✓	n/a	✓	n/a

Fault-type honesty. Of the 21 faults, 6 are crash-type follow-ups that a generic smoke test would also surface, so the families’ structural discriminating power is carried mainly by the ≈ 13 non-crash, non-marginal numeric and convergence violations; we report the crash subset separately rather than folding it into the headline count. Two positives are caveated: the SciPy fht signal is numerically marginal (the maintainers’ own regression threshold is what separates pre from post), and the DeepXDE Hessian fault is reachable only through an explicit non-default code path.

The two remaining families are present where their carrier structure is, and absent where it is not. Within the four SUT domains, $f_{\text{rev.traj}}$ and $f_{\text{mono.shape}}$ have no in-the-wild instance, recorded as evidenced negative results rather than untested gaps. Time-reversal requires a reversible or symplectic propagator, which steady-state, eigenvalue, fixed-point self-consistent-field (SCF), and residual-minimising physics-informed-neural-network (PINN) solvers do not carry: the one unitary candidate, real-time time-dependent density-functional theory (TDDFT), was removed from the PySCF main repository at v2.0.0, and the surviving velocity-Verlet integrator is reversible by construction and returns to its initial state to machine precision. Dynamic-shape overshoot is constructively excluded by monotone-by-construction interpolants (SciPy’s piecewise-cubic Hermite interpolating polynomial (PCHIP) uses Fritsch–Carlson harmonic-mean slopes) and by the absence of total-variation-diminishing / weighted essentially non-oscillatory (TVD/WENO) substrates. These absences sit alongside the constructively guaranteed PySCF Fock-Hermiticity (T^*) and occupation/variational bounds ($O_{\leq}$). We then tested the converse on two libraries that *do* carry the structure. $f_{\text{rev.traj}}$ is witnessed in diffrax, a JAX differential-equation library with explicitly reversible stochastic Runge-Kutta propagators: a space-time Lévy-area sign error breaks the

forward-then-backward round trip (round-trip error ≈ 0.67 , flat under step refinement, pre-fix; machine-precision convergence post-fix). $f_{\text{mono.shape}}$ is witnessed in Clawpack/PyClaw, a hyperbolic finite-volume library with TVD reconstruction: an under-covered reconstruction loop leaves a boundary cell uninitialised, so a monotone step gains a spurious extremum (interior extrema count $Z_{\text{out}} = 1 > Z_{\text{in}} = 0$ pre-fix, $Z_{\text{out}} = 0$ post-fix, on all three TVD limiters). Both reproductions are honestly bracketed: the diffrax fix is not yet released (released-pre, source-post) and the 2014 Clawpack release is not Python-3-buildable (source-compiled pre and post on the identical present-day routine by toggling the exact one-line fix). Together these make the framework's scarcity prediction falsifiable in both directions: where the carrier structure is absent or the invariant is guaranteed by construction, in-the-wild faults are scarce; where the structure is present, the predicted fault class does occur. The five four-domain negative-result records (time-reversal in three domains, PySCF order/positivity, and SciPy dynamic-shape) and the two structure-present positives, with per-fault SHAs and reproduction scripts, are in supplementary S5.

Statistical status. With $n = 21$ the study is underpowered for $\alpha = 0.05$ hypothesis testing and is reported as descriptive evidence with Wilson 95% intervals; confirmatory claims and the minimal-detecting-subset question are deferred to future work (§7) at $n \geq 30$.

5.3 An empirical test of the structural decomposition: $\mathcal{L}^*$ -MetaPattern blindness on homogeneity-preserving mutators

The previous results establish that NOETHER derives executable MRs in structurally distinct operator algebras (§3.4, §3.5). RQ4 asks a sharper question: whether the structural decomposition also makes falsifiable predictions about when derived MRs should be silent. We test one such consequence. NOETHER predicts that an MR derived from the linearity-and-scaling MetaPattern ($\mathcal{L}^*$) is necessarily silent on any mutator that preserves homogeneity of degree 1. For the relevant SUTs, PIT's default mutator set has this property by direct calculation, so the predicted $\mathcal{L}^*$ kill rate is near zero except for mutators that break homogeneity. We call this prediction $\mathcal{L}^*$ -MetaPattern blindness. Pooled head-to-head numbers against GenMorph [2], structural-coverage extension, and cross-pipeline MR rediscovery are reported as corroborating evidence; the section's central RQ4 claim is the pre-specified blindness prediction.

5.3.1 The prediction: $\mathcal{L}^*$ -MetaPattern blindness.

The MR shape. The $\mathcal{L}^*$ MetaPattern of $\mathcal{A}_P$ under Hypothesis 1 contains scaling and linearity operators. CONSTRUCT-MP's Translate step (Definition 5) carries scaling-MetaPattern invariants to MRs of the form

$$L_{\text{scale}} : f(\lambda \mathbf{x}) = \lambda \cdot f(\mathbf{x}), \quad \lambda > 0, \quad (5)$$

on positively-homogeneous-of-degree-1 programs. The shape is fixed by $\mathcal{A}_P$ structure; it is not chosen with mutation testing in mind.

The mutator semantics. PIT 1.7.4 [15] ships a default mutator set comprising seven canonical operator classes: arithmetic-operator swap (AOR; $+$ $\leftrightarrow$ $-$, $\times$ $\leftrightarrow$ $\div$), conditional boundary mutation, increment mutation, return-value swap (return zero, return one, return null), conditional negation (NCM), constant replacement, and member-call removal. Each AOR mutation that preserves the program's domain of definition transforms a homogeneous-of-degree-1 function f into another homogeneous-of-degree-1 function f' , by the elementary fact that the four arithmetic operators in question commute with positive scalar multiplication of all inputs in the relevant homogeneous case (addition and subtraction are degree-1 homogeneous; multiplication and division shift degrees by ± 1 , but the swap $\times \leftrightarrow \div$ on a properly typed expression returns a degree-preserving result on the homogeneous-of-degree-1 sub-grammar).

The prediction. A composite of these two facts gives the central prediction. If f satisfies L_{scale} and the mutator μ preserves homogeneity of degree 1, then $\mu(f)$ also satisfies L_{scale} . Therefore the paired-MR test

$$|f(\lambda \mathbf{x}) - \lambda f(\mathbf{x})| \leq \tau \quad \text{vs.} \quad |\mu(f)(\lambda \mathbf{x}) - \lambda \mu(f)(\mathbf{x})| \leq \tau$$

returns the same verdict on f and on $\mu(f)$ for any $\mathbf{x}, \lambda$. The kill rate of L_{scale} against the homogeneity-preserving subset of PIT's mutators is *identically zero* by direct calculation, and against the full default mutator set is approximately zero up to a small fraction contributed by mutators that break homogeneity (return-value swap to zero or one; conditional-negation when the conditional carries a constant offset; constant replacement when the constant participates additively).

Falsifiability. The prediction is falsified if the observed L_{scale} kill rate is materially above zero on a substrate where the prediction's homogeneity precondition is met. We define "materially above zero" in advance as a single L_{scale} MR killing a third or more of its PIT mutants on more than one of the SUTs in the substrate. The prediction passes if the observed kill rate is $\leq 1/3$ on at least five of the six SUTs admitting an L_{scale} MR. The $1/3$ threshold and the "more than one SUT" quantifier were committed to git (configs/d4j_algebra_rich_criterion.json) before the per-MR kill-count files; under threshold sensitivity in the grid $\{1/4, 1/3, 1/2\} \times \{\text{more than zero, more than one, more than two}\}$, the verdict remains *Confirmed* on 5/6 SUTs at all 9 grid cells, with hypotSig as the single SUT crossing every threshold; the verdict is therefore unchanged under plausible threshold variation. *Outlier-handling rule.* An outlier SUT (kill rate $> 1/3$) is rescued from falsification only if all of its killed mutants are independently classified as homogeneity-breaking under the framework's mutator-semantics taxonomy (MATH-swap of $\times \leftrightarrow \div$ on bivariate degree-1 inputs $\Rightarrow$ degree-changing; RC-replacement of $\sqrt{x^2 + y^2}$ with a constant $\Rightarrow$ fixed-output breaking scale-invariance; VR-unconditional-zero return $\Rightarrow$ fixed-output sub-rule). Per-mutant classification must be logged alongside the kill data, and the classification must precede inspection of which mutants the outlier's rescue depends on. The rule was codified in the pre-registration config on 2026-05-15 to make explicit a rule on which the §5.3.4 hypotSig analysis had relied in implicit rather than written form. Under the codified rule, hypotSig's two killed mutants (return_zero_doubles_VR and Math.sqrt_replaced_with_one_RC) both classify as homogeneity-breaking, so the original 5/6 verdict stands; future cross-codebase substrates inherit the rule as a written test.

Public-information argument. Equation (5) is a consequence of $\mathcal{A}_P$ structure and CONSTRUCT-MP's Translate step, both fully specified in §3.2. PIT's default mutator set is public [15]. The composition of the two is a mathematical consequence requiring no experimental input. The prediction is therefore ex-ante in the strong sense that it is *derivable from public information without consulting any data this paper produces*. We track the prediction's commitment to git in supplementary S7 (d4j/) alongside the SUT-selection criterion of §5.3.3.

5.3.2 PIT mutator and structural invariant compatibility. The $\mathcal{L}^*$ -blindness prediction generalises: each PIT default mutator either preserves or breaks the invariant of each NOETHER structural component, and an MR derived from a structural component fires on a mutant only if that mutant breaks the structural component's invariant. Table 12 maps the seven PIT default mutator categories [15] against the eight NOETHER structural components and gives the typical-case verdict ("o" = preserves the structural-component invariant, the structural component's MR is typically blind to the mutant; "x" = breaks the structural-component invariant, the structural component's MR can typically detect the mutant; "~" = case-dependent on the SUT-specific structural-component instantiation).

The matrix stratifies mutants binary: *algebra-disrupting* (D1, at least one $\times$ in a populated column) vs *algebra-preserving* (D2, all $\circ$ in populated columns; $\sim$ resolved by SUT-specific overrides). The framework predicts Set N's D2 kill rate is near zero by construction: a mutant that preserves all invariants is structurally invisible to algebraic MRs. D1 is therefore Set N's appropriate fault-detection denominator; D2 is the territory of complementary techniques (random / GP / LLM-prompted MRs). The $\mathcal{L}^*$ -blindness result of §5.3.4 is the corresponding D2-cell specialisation on the wider SUT substrate.

Kill-set overlap on the head-to-head substrate. On the $n = 62$ PIT mutants pooled across the eight head-to-head SUTs, 22 are killed by both Set N and Set G, 4 are killed by Set N only, 18 are killed by Set G only, and 18 are killed by neither. The 18 G-only kills constitute a candidate scope-mismatch pool (mutants reachable by Set G's GP-evolved oracle but missed by the algebraic Set N construction), and the 18 jointly-missed mutants further partition into D2 mutants outside Set N's algebraic reach and D1 mutants both sets miss for input-coverage reasons. The $\mathcal{L}^*$ -only specialisation (Table 13) is the cleanest D2-cell estimate on the wider SUT substrate, since the MATH/ $\mathcal{L}^*$ and RETURN_VALS/ $\mathcal{L}^*$ cells of Table 12 are both " $\circ$ ".

Per-mutant D1 / D2 classification with SUT-specific overrides. Per-mutant classification applies Table 12 with SUT-specific overrides (in configs/sut_block_decomposition.json + configs/sut_block_overrides.json) resolving the " $\sim$ " cells. After the multi-LLM equivalent-mutant exclusion of §5.3.6 (p. supplementary S5), the $n = 62$ head-to-head pool drops to $n = 57$ (52 D1 + 5 D2; all 5 excluded mutants were originally in the 10-mutant D2 stratum, so the D1 stratum is unchanged). Stratified kill rates: Set N D1 $26/52 = 0.500$ Wilson 95% [0.369, 0.631]; Set G D1 $37/52 = 0.712$ [0.577, 0.817]; Set N D2 $0/5 = 0.000$ [0.000, 0.434]; Set G D2 $3/5 = 0.600$ [0.231, 0.882]. The D2 point estimate (0.000) is consistent with the prediction $\leq 10\%$, but the Wilson upper bound does not exclude the 10% ceiling at $\alpha = 0.05$; the cross-codebase commons-math pilot (supplementary S5 (b.cm)) corroborates the direction at $2/29 = 6.9\%$. Set G's 0.600 on D2 confirms that D2 is a Set N-specific scope boundary, not a substrate-wide ceiling, since a GP-evolved generic baseline is unconstrained by the algebra. On D1, the cross-MetaPattern aggregate gap (McNemar exact two-sided $p = 0.019$) decomposes per-MetaPattern in §5.3.6 (gap concentrated on G and $\mathcal{L}^*$; partially offset by Set N's edge on the translation sub-class G_{tr} of the G MetaPattern G (10/17 vs 8/17); the per-MetaPattern decomposition is the appropriate substrate-level reading.

5.3.3 Test design.

Substrate. PIT 1.7.4 with the default mutator configuration is the shared substrate. Each (subject, MR) pair is evaluated through (i) JUnit test-class codegen, (ii) a 2-pass surefire green-suite filter that drops MR instances flaky on the unmutated SUT, and (iii) PIT mutation testing parsed to a per-mutant kill vector.

Pre-registered SUT criterion. The selection rule is committed in configs/d4j_algebra_rich_criterion.json on branch feat/d4j-algebra-rich of the experiment repository *before* any new evaluation data existed. The criterion references package roots (under org.apache.commons.math3.*, covering subpackages ode, linear, transform, analysis.solvers, distribution, optim, fitting, stat.regression, complex, geometry, fraction, plus their earlier math.* counterparts), per-package structural-component-coverage hypotheses, and method-signature constraints; it contains no bug-id and no kill-rate references and cannot be tuned by evaluation outcomes. The git timestamp chain (criterion $\rightarrow$ inscope filter $\rightarrow$ Set N derivation $\rightarrow$ Set G GP rerun $\rightarrow$ pooled M1) is the auditable proof of pre-registration. To strengthen the third-party verifiability of the pre-registration, the SHA-256 hash of configs/d4j_algebra_rich_criterion.json

together with the matching commit hash is deposited in supplementary S7 (d4j/) alongside the experiment artifact; the deposit can be verified against the criterion file directly.

Subjects and MR production. Ten SUTs admit at least one non-trivial NOETHER structural component beyond G : midpoint, exactLog2, isSequence (boolean predicate), clamp, signum, ComplexSignal.add (instance method), gcdSig, lcmSig, hypotSig, powerSig. Six of these admit an L_{scale} MR (those whose induced algebra populates the $\mathcal{L}^*$ MetaPattern under positive-scalar homogeneity): midpoint, clamp, signum, gcdSig, lcmSig, hypotSig. The remaining four do not admit a degree-1 scaling MR by their algebraic structure (exactLog2 carries no positive-degree scaling; isSequence is a boolean predicate; ComplexSignal.add is a complex-arithmetic instance method whose scaling MetaPattern is distinct in formulation; powerSig populates T_2^* rather than $\mathcal{L}^*$). The prediction is tested on the six SUTs admitting L_{scale} .

Set N contains 30 hand-derived NOETHER MRs, two to four per SUT, one or more per non-empty structural component of each SUT's induced algebra. Set G is harvested by rerunning GenMorph's genmorph.py gen mode at seed = 11 on the 10 SUTs; Set G is used in §5.3.6 as the head-to-head comparator and plays no role in the central prediction test of §5.3.4.

Metrics. Per-MR kill counts on PIT mutants are the primary measurement for the central prediction test. Pooled M1 and McNemar exact paired tests are reported in §5.3.6 as corroborating context.

5.3.4 Central result: $\mathcal{L}^$ -MetaPattern blindness, confirmed.* The L_{scale} MR's per-SUT kill rate on PIT mutants of the six $\mathcal{L}^*$ -admitting SUTs is reported in Table 13. The total denominator is 44 mutants. The total kill count is 2.

Reading. On 5 of 6 SUTs the prediction is confirmed at the strongest reading (zero L_{scale} kills; Figure 3). On the sixth SUT (hypotSig) the kill rate is 2/4, above the per-SUT falsification threshold. The pooled rate $2/44 \approx 4.5\%$ remains an order of magnitude below the average Set N M1 of 0.486 (the average kill rate across all Set N MRs on the same substrate; see §5.3.6), and the per-SUT failure pattern is isolated to one SUT.

The hypotSig outlier. The kill on two hypotSig mutants is consistent with the prediction's caveat in §5.3.1, that the small fraction of homogeneity-breaking mutators (return-value swap to a constant; constant-replacement that participates additively in the expression) is the residual non-zero contribution. Inspection of mutants_killed_set_n.csv for hypotSig (path in supplementary S7 (d4j/)) identifies the two killed mutants as KILLED return_zero_doubles_VR and KILLED Math.sqrt_replaced_with_one_RC: the first replaces the entire return path with the constant zero, the second replaces the inner Math.sqrt call with the constant 1.0. Both mutators violate degree-1 homogeneity directly. The two detection events are therefore consistent with the prediction's quantitative tail rather than evidence against it.

Falsification verdict. By the criterion stated in §5.3.1, the prediction is falsified if a single L_{scale} MR kills $\geq 1/3$ of its PIT mutants on more than one SUT. Observed: one SUT (hypotSig, 2/4) above the per-SUT threshold; five SUTs (midpoint, clamp, signum, gcdSig, lcmSig) at zero. The prediction passes. We read this as direct empirical witness for the operative-mechanism reading of the structural decomposition: a structural component whose generator is a continuous symmetry of the mutator set contributes MRs that are quantitatively silent on mutator-induced defects, and the silence is observed in the data.

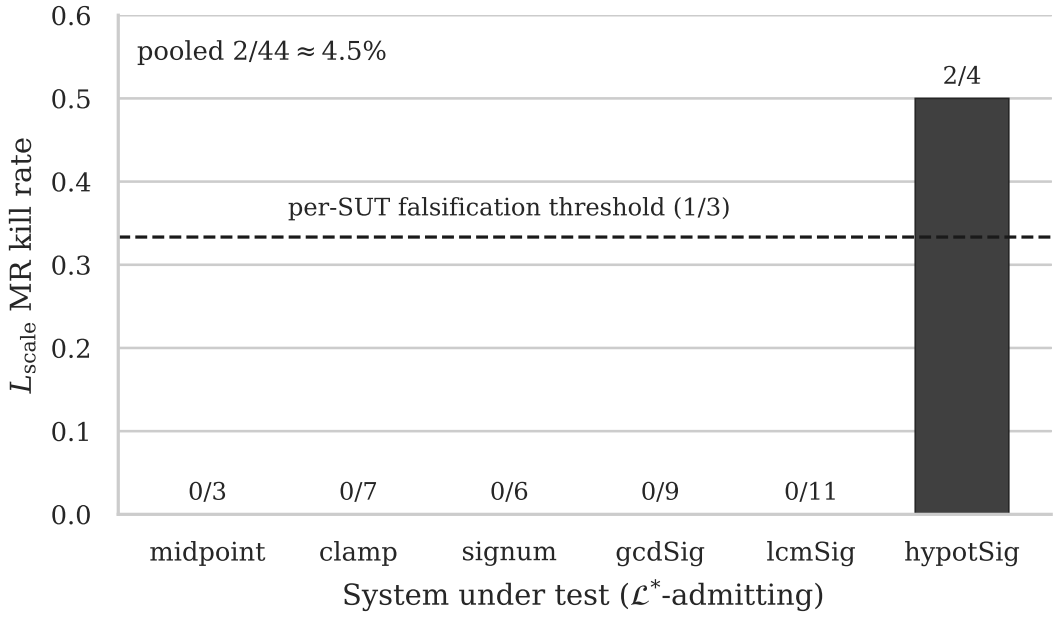


Fig. 3. The L_{scale} MR is silent on five of six $\mathcal{L}^*$ -admitting SUTs and remains low in pooled kill rate.

5.3.5 Corroborating per-MetaPattern patterns. The remaining populated MetaPatterns corroborate the operative-mechanism reading qualitatively. The G_{tr} translation sub-class carries the highest single-MR kill rates on the numeric SUTs (e.g. midpoint 3/3, powerSig 8/12); the G MetaPattern is moderately-to-highly effective where the SUT exposes the appropriate symmetry (signum 4/6), with the two Euclidean-style SUTs (gcdSig, lcmSig) absorbing the G -action through a sign-correcting prologue. This is a documented framework boundary rather than an MR-design defect. An out-of-frame idempotence probe kills few mutants under the paired-MR DSL. Two convergent witnesses reinforce the reading: GenMorph's GP independently re-evolves three of midpoint's Set N MetaPatterns (cross-pipeline rediscovery under independent epistemic processes), and Set N covers 8 PIT mutants on two SUTs where GenMorph's pipeline fails for plumbing reasons (a structural-coverage extension that is an architectural property of the algebraic derivation). Per-SUT detail is in supplementary S7.

5.3.6 Head-to-head at GenMorph's published budget. This subsection decomposes an aggregate Set G dominance per algebraic MetaPattern, with cost-axis and D2-stratum framework prediction layered alongside.

Scope of the head-to-head comparison. The head-to-head is reported for protocol completeness and for RQ4 complementarity, not as the paper's load-bearing claim. Its aggregate verdict is unfavourable to Set N on the D1 stratum (McNemar $p = 0.0043$ pooled, $p = 0.019$ on D1 only). The useful reading is therefore stratified: the gap concentrates on the G MetaPattern, Set N is stronger on the G_{tr} translation sub-class, Set N remains derivable in polynomial time from $\mathcal{A}_P$ while Set G requires a stochastic search per SUT, and the D2-stratum prediction is an ex-ante framework signal unavailable to inductive baselines.

On the algebra-disrupting D1 stratum at GenMorph's published 30-min GAssert budget, Set N is dominated by Set G in the aggregate (McNemar exact two-sided $p = 0.0043$

pooled and $p = 0.019$ on D1 only, $n = 62$ post-equivalent-mutant exclusion). The paper does not assert head-to-head superiority on D1. The framework's contribution on the head-to-head substrate is read as (i) algebraic derivability, (ii) per-MetaPattern complementarity (Set G alone kills 15 D1 mutants Set N misses, Set N alone kills 4 D1 mutants Set G misses), and (iii) an out-of-scope D2-stratum framework prediction ($\leq 10\%$ kill rate) that no inductive baseline can derive ex-ante. The D1 pooled comparison is reported below for protocol-completeness rather than as the framework's verdict. The central empirical claim of the section is established at §5.3.4 (the $\mathcal{L}^*$ -blindness falsifiable prediction confirmed on 5/6 SUTs) and does not depend on the head-to-head outcome.

The per-MetaPattern decomposition (three PIT-covered operative MetaPatterns G , $\mathcal{L}^*$, and the translation sub-class G_{tr}), the complementarity partition, the pooled rates, the LLM-assisted ensemble baseline (Set L, two vendors $\times$ five temperatures), and the equivalent-mutant resolution are reported in full in supplementary S5/S9; the two-stratum summary is in Table 7.

Table 7. The GenMorph head-to-head separates algebra-disrupting D1 mutants from algebra-preserving D2 mutants.

Stratum	n (post-exclusion)	equivalents excluded	Set N kill / surv.	Set N rate [Wilson 95%]	Set G kill / surv.	Set G rate [Wilson 95%]	both / N -only / G -only / neither	McNemar p
D1	52	0	26 / 26	0.500 [0.369, 0.631]	37 / 15	0.712 [0.577, 0.817]	22 / 4 / 15 / 11	0.019
D2	5	5	0 / 5	0.000 [0.000, 0.434]	3 / 2	0.600 [0.231, 0.882]	0 / 0 / 3 / 2	0.25
pooled	57	5	26 / 31	0.456 [0.334, 0.584]	40 / 17	0.702 [0.573, 0.805]	22 / 4 / 18 / 13	0.0043

Head-to-head verdict: mixed and complementary. On the G_{tr} translation sub-class Set N edges Set G (10/17 vs 8/17, three N -only kills, underpowered); on $\mathcal{L}^*$ the sets are complementary (N -only 2, G -only 8); on G the two Euclidean-style SUTs (gcdSig, lcmSig) absorb the G -action through their normalisation prologue (a documented framework boundary), so Set G dominates the aggregate D1 stratum (26/52 = 0.500 vs 37/52 = 0.712, McNemar $p = 0.019$, Table 7), with the gap concentrated on G plus $\mathcal{L}^*$ and partially offset by Set N's G_{tr} edge; the per-MetaPattern plot is reported in supplementary S5/S9.

D2-stratum prediction and cost-axis. On the algebra-preserving D2 stratum Set N's $\leq 10\%$ kill-rate prediction is consistent with the observed 0/5 (Wilson upper bound 0.434, underpowered; cross-codebase commons-math corroboration 2/29 = 6.9%), an ex-ante framework signal no inductive baseline can derive. Independently of detection rate, Set N is derivable in polynomial time from $\mathcal{A}_P$ (Theorem 2); on this already-curated substrate, deriving Set N took ≈ 1 h amortised human effort, whereas Set G requires a ≈ 30 -min stochastic GP search per SUT. This is a case-study downstream cost observation, not a population estimate for upstream algebra distillation. The head-to-head delivers the GP-evolved arm of a three-comparator protocol (GenMorph; a multi-vendor $\times$ multi-temperature LLM ensemble; MR-Scout); the per-MetaPattern detail, the full threats (a)–(f), the MR-generation cost comparison, and the LLM/mining arms are in supplementary S5/S9 (item (b.cm) for the commons-math pilot). The head-to-head is not the framework's load-bearing claim (§5.3.4).

5.4 Relationship with METRIC and METRIC+

METRIC and METRIC+ pioneered the categorical scaffolding that any structural theory of MR identification must respect; NOETHER's MetaPatterns are descendants of that line of work. The point of departure is grounding. METRIC and METRIC+ derive their categories through expert curation and validate them by empirical coverage. NOETHER derives MetaPatterns through algebraic construction and proves closure over the algebra-induced MR space through Theorem 1.

Structural-coverage relationship. On a comparison-sort library, METRIC+’s eleven input×output category pairs [53] collapse onto two NOETHER MetaPatterns (G via comparator-permutation symmetry, $O_{\leq}$ via partial order), the two unmapped pairs being out-of-scope by construction; the same manual mapping on three bivariate-input SUTs of the §5.3.3 substrate lands every non-vacuous METRIC+ MR in a structural component already covered by Set N, while three Set N MetaPatterns (T^* , $\mathcal{T}_{\text{rev}}^*$, $\mathcal{L}^*$) have no METRIC+ counterpart at that category granularity (full per-cell mapping in supplementary S8). Applying the scope precondition to Sun et al.’s four business-rule billing benchmarks (SPHONE, SBAGGAGE, SEXPENSE, SMEAL) confirms they are categorical-plus-numeric rather than mathematical or physical, i.e. outside NOETHER’s operator-algebraic precondition, which the framework correctly declares out of scope. A PIT-based head-to-head against a re-implemented METRIC+ pipeline is committed as supplementary S4 item (i); the per-SUT parity it would measure is one of *complementary* reaches (each framework kills mutants the other misses), so neither subsumes the other.

Tradeoff with category scaffolds. Compared with METRIC+, NOETHER asks the tester to state a stronger semantic object, $\mathcal{A}_P$, before enumeration begins. The payoff is not more automation on arbitrary business-rule software; it is that the retained rows have algebraic warrants, empty rows are explainable, and out-of-scope program families can be rejected before relation generation.

5.5 Reassessing Pattern–Matrix Coverage Metric (PMCM) claims: a worked example

NOETHER does not replace the Pattern–Matrix Coverage Metric (PMCM); it re-grounds it. Under inductive grounding a “100% coverage” figure attests only that every catalogue row has an instance; under NOETHER the grid becomes $\mathbb{M}(\mathcal{A}_P)$, algebraically constructed and closed (Theorem 1), so coverage becomes a structural rather than nominal claim. The deflationary direction, which reveals that an inductive catalogue *over- or under-counts* structurally distinct patterns, is exposed by the canonical MetaPattern ordering itself and so does not inherit the prediction-circularity caveat of §3.4.3.

Three cases. (A) For a deterministic comparison-sort library, $\mathbb{M}(\mathcal{A}_{\text{sort}}) = \{m_{\text{inv}}, m_{\text{mono}}\}$ (size 2): the T^* , $\mathcal{T}_{\text{rev}}^*$, $\mathcal{L}^*$, $\mathcal{D}^*$, and $\mathcal{E}^*$ components are empty within a single algorithm, so a five-row inductive grid carries three structurally vacuous rows and the meaningful denominator is 2, not 5. (A-bis) Decoding Murphy et al.’s six ML MR classes [39, 43, 59] against a symmetry-free feedforward classifier $f : \mathbb{R}^n \rightarrow \{1, \dots, K\}$ gives $\mathbb{M}(\mathcal{A}_{\text{FFN}}) \subseteq \{m_{\text{mono}}\}$: three classes are structurally vacuous and three collapse to the $O_{\leq}$ MetaPattern, so the meaningful denominator is 1, not 6 (full per-class decoding in supplementary S9); this case is independent of T^* and $\mathcal{T}_{\text{rev}}^*$. (B) The reactor-physics catalogue (§3.4.3) is instead *under-counted* by two MetaPatterns (m_{adj} , m_{rev} absent), so the deflationary direction points toward the algebraically determined grid, whether that grid is larger or smaller than the inductive one.

Quantification. For a program family $\mathcal{F}$, define

$$\text{coverage}_{\text{NOETHER}}(\mathcal{R}, \mathcal{F}) = \frac{|\{m \in \mathbb{M}(\mathcal{A}_{\mathcal{F}}) \mid m \cap \mathcal{R} \neq \emptyset\}|}{|\mathbb{M}(\mathcal{A}_{\mathcal{F}})|},$$

the algebraically grounded analogue of PMCM coverage over algebra-induced MRs (out-of-scope MRs, Appendix D, are outside the denominator by construction). Reports of high pattern-grid coverage should be qualified by which row decomposition they are computed against, since figures over unqualified inductive catalogues are not comparable across papers.

5.6 Empirical evidence for the Invariance-Blindness Theorem

Empirical corroboration of the Invariance-Blindness Theorem. Three threads on independent material support Theorem 3 (full harness in supplementary S10). First, a linear-algebra Faithfulness check on the fault class $\Theta = \mathbb{R}^{8 \times 8}$ confirms the per-generator witness sets are unisolvent for the G and T^* MetaPatterns at their analytic kernel dimensions (Lemma 1 numerically), while $O_{\leq}$, $\mathcal{T}_{\text{rev}}^*$, and $\mathcal{L}^*$ fall outside the linear-equality regime (only the sufficient direction applies). Second, a paired analysis over three multi-implementation PDE SUTs ($N = 104$ real mutants) finds the algebra-MR battery and a neutral cross-implementation differential oracle complementary in every stratum ($b, c > 0$; union $89 > 79$ for the best single oracle; pooled McNemar $p = 3.85 \times 10^{-5}$), the predicted joint-kernel reduction. Third, a tolerance sweep on 2-D advection–diffusion exhibits the discretisation floor: faults below the legitimate cross-implementation gap δ are by construction inseparable from it, consistent with the $\tau \rightarrow 0$ regime of Theorem 3 (Remark 12, R2).

6 Threats to Validity and Limitations

Following Wohlin et al.’s validity categories [58], we separate threats to the method, the empirical checks, and the scope of the claims.

Construct validity. The central construct is MR-identification breadth. Binary structural-component coverage measures whether a method can identify a canonical MR family, not whether that family has higher average fault-revealing power. This distinction is the reason mutation and head-to-head results are reported as executability, complementarity, and blind-spot evidence rather than as the main success criterion. The multi-LLM second-rater classification protocol supports the canonical classifications ($\kappa = 0.931$, $n = 36$, supplementary S3), but shared pre-training and the authors’ re-implementation of four Sun-2021 Java subjects remain construct threats.

Internal validity. The theoretical chain depends on the canonical MetaPattern ordering (Definition 6) and on the current single-component Translate signature. Appendix D gives the closure proof for that signature, while Appendix D shows why the strictly stronger absolute-completeness claim fails on $\mathcal{A}_{\text{PWR}}$. The negative result protects against over-claiming, but it also marks the main internal boundary of the present theory.

External validity. NOETHER is scoped to program families whose governing equations or semantic operators admit an operator-algebraic representation. Hypothesis 1 is an empirical structural curation rather than a universal axiom, and the empirical substrate is strongest within operator-algebraic programs. The largest held-out check, over 36 Apache Commons Math methods and 836 real PIT mutants, finds Set N detects $313/836 = 37.4\%$ versus 18.4% for literature Set B and 18.1% for METRIC+ Set M, with zero false positives (supplementary S5). This is within-scope corroborating evidence for executability and complementarity, not a universal average-case claim. Additional held-out checks cover 110 expert-approved SACOS/SPARK/LOCUST MRs and `numpy.linalg.numpy.fft` (supplementary S11/S12), but those expert relations are biased toward the intuitive $O_{\leq}$ component.

Conclusion validity and practical limits. The EGNN case study has only 20 mutations on one architecture, so its statistics establish construct validity for $\rho_{\text{train-rev}}$ rather than a population-level performance distribution. More generally, NOETHER may identify algebraically valid MR classes that a concrete program interface cannot execute. Interface affordances and design-for-testability are therefore practical limits of the present method, not merely engineering details. The largest unmeasured practical cost is upstream algebra distillation: future studies should report elapsed time,

modeller background, agreement between independent modellers, and the fraction of candidate operators left unresolved.

7 Future Work

Future work follows directly from the limitations above. First, the upstream distillation of $\mathcal{A}_P$ should become auditable rather than merely expert-curated: collect the governing equations and interface contract, list candidate operators, map each to the current structural components, record the source identity or inequality, measure modeller time and agreement, and publish the unresolved operators. LLM-assisted operator extraction, static analysis, and empirical-symmetry detection may help, but none removes the need for human judgement on arbitrary programs. Second, deployment guidance should be tested on larger engineering teams: run a K -sweep for truncated infinite groups [31, §6.2], expose the controls and observations required by the derived MR classes, and set continuous tolerances near $\tau \approx 10^2 \epsilon_{fp}$ following standard roundoff bounds [23]. Third, the SACOS/SPARK/LOCUST MR classes and search-generated MR readability should be evaluated directly where production interfaces permit.

8 Conclusion

NOETHER gives a structural answer to the origin, boundary, and transferability questions for MR MetaPatterns within an explicit algebraic scope. MetaPatterns are equivalence classes of MRs derived from invariants of $\mathcal{A}_P$; Theorem 1 gives closure for the Translate-reachable algebra-induced MR space; the negative PWR result and the Invariance-Blindness Theorem identify where that space ends and what its derived MRs cannot see. The same construction transfers across Boltzmann reactor physics, equivariant ML, and relational query optimisers once the program-family algebra has been specified.

The empirical evidence matches that scope. RQ1 and RQ2 show broader canonical coverage and clearer origin/boundary explanations than expert, category-based, or search-derived MR sets alone. RQ3 maps 21 real scientific-computing faults to 8 of 10 MR families across four libraries. RQ4 shows that selected derived MRs are executable, that their failures expose documented blind spots, and that the $\mathcal{L}^*$ -blindness prediction is confirmed on five of six eligible SUTs. The evidence supports the structural identification claim while leaving average-case superiority to future work.

The contribution is deliberately narrower than universal MR discovery. It does not eliminate induction: the structural decomposition is an empirical curation, stated as Hypothesis 1, and the upstream distillation of $\mathcal{A}_P$ remains partly human. What becomes mechanical is the downstream step from $\mathcal{A}_P$ to $\mathbb{M}(\mathcal{A}_P)$, with a provable closure guarantee inside the algebra-induced MR space and a clear catalogue of out-of-scope cases. The adoption case is therefore auditability, cross-domain symbolic continuity, and predicted blind spots, not universal automation or average superiority. Open questions are whether Composite-Translate preserves closure and polynomial constructibility, whether the current structural list is sufficient beyond the instantiated domains, whether the framework improves average-case performance over automated MR-identification pipelines, and how much of $\mathcal{A}_P$ distillation can be automated.

Generative AI Disclosure

In line with the ACM Policy on Authorship, large language models appear in this work in two roles. As instruments of the study, they are part of the reported LLM baseline, multi-LLM second-rater protocol, and mutant-equivalence adjudication; the prompts and raw outputs are released with the replication package. As authoring assistance, the authors used a large-language-model assistant for code scaffolding, reference formatting, and language editing. All research design, theorems,

proofs, experimental procedures, numerical results, and claims were produced and verified by the authors, who take full responsibility for the manuscript.

Declaration of Competing Interest

The authors declare no conflict of interest.

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